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Characterisation, Field Modelling, and Radiation Testing of a Commercial Tunnelling Magnetoresistance (TMR) Sensor for CubeSat Attitude Determination and Control

Final Year Technical Report

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[TODO: insert submission date]

Abstract

Tunnelling magnetoresistance (TMR) sensors offer chip-scale, high-sensitivity magnetic field detection that is attractive for CubeSat attitude determination and control, yet the technology has little space heritage, and little device-level data exists on how a commercial part behaves when it is saturated by a strong field or after irradiation. This report characterises a low-cost commercial TMR sensor, the CT100, using two custom printed circuit boards: a standalone board exposing the raw bridge output, and an integrated board that adds an STM32 microcontroller and an independent current-sense reference to drive and log a controlled test field from a PCB trace. A field model built from the board stack-up and the datasheet position of the sensing element predicts a current-to-output gradient of $1.33\,\mathrm{mV\,A^{-1}}$; all nine fitted sweeps lie within -5% to $+9\%$ of this prediction, equivalent to a sensitivity of 4.3 to $4.9\,\mathrm{mV\,V^{-1}\,mT^{-1}}$ against a typical datasheet value of 4.5. The saturation ceiling was within about 5% of $V_{\mathrm{DD}}/3$ on every unit and symmetric about zero, and the output recovered from saturation in 40 to 85 ms as an exponential decay with a time constant of about 9 to 14 ms. At a 10 Hz magnetometer rate this requires the first one to two samples after a saturating event to be discarded. On a fresh sensor, saturation shifted the zero-field output by about 16 mV (roughly 1 mT) while the gradient changed by only 0.5% and the response stayed linear, so the offset must be re-estimated after saturation. After a total ionising dose of 40.76 krad(Si), no change attributable to radiation was found in the ceiling, recovery or gradient, and ferrite shielding produced no measurable change. The results suggest that the CT100 is a promising low-cost COTS candidate for CubeSat magnetometry: its intrinsic noise is low enough to resolve the geomagnetic field, but the unamplified 12-bit readout used here is not, so a gain stage or a higher-resolution converter is required.

Signed Statement of Contribution

This project was carried out jointly by Sam Ye and Daniel Tieu under the supervision of Dariusz Kacprzak. Sam Ye was responsible for **[TODO: list: sensor characterisation, design of both PCBs, firmware, field modelling and Ansys simulation, saturation, radiation, shielding and hysteresis testing]**. Daniel Tieu was responsible for **[TODO: list: thermal-cycling and temperature-sensitivity testing]**.

Signed: _____ Date: _____

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Glossary of Terms

ADCS	Attitude Determination and Control System
ADC	Analogue-to-Digital Converter
AMR	Anisotropic Magnetoresistance
COTS	Commercial-Off-The-Shelf
EPS	Electrical Power System
FEM	Finite Element Method
FGM	Fluxgate Magnetometer
GMR	Giant Magnetoresistance
HAL	Hardware Abstraction Layer (STM32 firmware library)
I^2C	Inter-Integrated Circuit (serial communication bus)
INA	Current-sense amplifier (current monitor IC)
krad(Si), Mrad(Si)	Kilorad / megarad of absorbed dose in silicon – units of Total Ionising Dose
LDO	Low-Dropout (linear voltage regulator)
MCU	Microcontroller Unit
MgO	Magnesium Oxide (crystalline tunnel-barrier material)
MI	Magneto-Inductive (sensor technology)
MTJ	Magnetic Tunnel Junction
OBC	On-Board Computer
PCB	Printed Circuit Board
SWaP-C	Size, Weight, Power, and Cost
SWV	Serial Wire Viewer (ARM Cortex debug trace interface)
TID	Total Ionising Dose
TMR	Tunnelling Magnetoresistance
TRL	Technology Readiness Level
Rad-Hard	Radiation-Hardened

1 Introduction

CubeSats are nanosatellites (1–10 kg) built from standardised 10 cm cubic units, offering low-cost, rapid access to space relative to traditional monolithic satellite missions. This miniaturisation is achieved at the expense of severe Size, Weight, Power, and Cost (SWaP-C) constraints, which particularly affect the Attitude Determination and Control System (ADCS): the subsystem responsible for orienting a CubeSat for communications, power generation, and payload pointing. Onboard magnetometers are the primary means by which a CubeSat senses its attitude relative to the Earth's magnetic field, so the choice of magnetometer technology is central to ADCS performance within a tightly constrained mass, volume, and power budget.

Heritage fluxgate magnetometers offer excellent sensitivity but are too large and power-hungry for CubeSat platforms, motivating a shift toward chip-scale magnetoresistive sensors. Among these, Tunnelling Magnetoresistance (TMR) sensors offer higher sensitivity than AMR or GMR sensors in a footprint of only a few millimetres square. However, TMR remains comparatively immature for space use – typically cited at Technology Readiness Level (TRL) 3 – with comparatively little published characterisation data under space-relevant stresses such as strong transient fields from onboard actuators (e.g. magnetorquers) and ionising radiation.

This report documents the design, construction and testing of a project evaluating a commercial-off-the-shelf (COTS) TMR sensor, the CT100, for CubeSat ADCS/EPS applications. Two custom PCBs were developed: a standalone board exposing the raw bridge output, and an integrated telemetry board combining the sensor with microcontroller-based logging and an independent current-based field reference. The research question, objectives and hypotheses that structure the work are set out in Section 3.

A parallel thermal-cycling test campaign investigating TMR temperature sensitivity and thermal-shock response was led by project partner Daniel Tieu and is reported separately; it is outside the scope of this report.

The remainder of this report follows the project's own design and testing sequence. Section 2 reviews the literature and identifies the research gap, and Section 3 states the research question, objectives and hypotheses. Section 4 describes the CT100, Section 5 the instrumentation, and Section 6 the standalone board and the sensor-only saturation tests. Section 7 covers the integrated board, the field model and the firmware. Section 8 presents the non-irradiated saturation recovery and current sweeps and validates the instruments against each other. Sections 9, 10 and 11 cover radiation, shielding and the effect of saturation on the sensor's offset. Sections 12 and 13 discuss the findings and conclude.

2 Literature Review

2.1 Magnetometer Technologies for CubeSat ADCS

Fluxgate magnetometers (FGMs) have been used in space exploration for decades: an alternating current drives a coil-wrapped permeable core toward saturation, and an external field produces a measurable imbalance between drive and sense coil currents. FGMs offer excellent sensitivity (of order $10\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$) but are far too large and power-hungry for CubeSat platforms, typically weighing over 1 kg and drawing more than 10 W [1, 2]. Magneto-inductive (MI) sensors, which sense an external field through the change it induces in the resonant frequency of

an LR oscillator, trade sensitivity for a much smaller footprint and lower power draw [1].

Hall-effect sensors are inexpensive and, because they contain no ferromagnetic material, immune to magnetic saturation. However, their comparatively low sensitivity confines them to detecting fields above roughly 1 mT, and while flux concentrators can improve their effective sensitivity, doing so reintroduces the mass and volume penalty that SWaP-C constraints are meant to avoid [3]. Magnetoresistive sensors – AMR, GMR, and TMR – instead occupy the sub-milliTesla sensitivity range needed for geomagnetic-field sensing while remaining chip-scale. AMR sensors (typically permalloy thin films) achieve high resolution and bandwidth but saturate above roughly 1 mT and often require periodic magnetic reset [3]. GMR sensors, based on a multilayer stack of alternating ferromagnetic and non-magnetic conductive layers, offer higher sensitivity again but are more prone to hysteresis-induced offset and can be permanently degraded by strong fields [3]. Chip-scale magnetoresistive sensing has already reached flight heritage: the AMR-based MAGIC instrument, for example, has demonstrated a 104 g, 0.5 W dual-sensor vector magnetometer in orbit on the RadCube CubeSat [4].

TMR sensors extend this trend further. They are built from magnetic tunnel junctions, whose spin-dependent tunnelling through an ultra-thin barrier was first demonstrated by Jullière [5] and made practical at room temperature with crystalline MgO barriers [6, 7]. TMR is the newest of the magnetoresistive technologies and is generally cited at TRL 3, compared with TRL 9 for AMR and GMR [8]. The specific device used in this project is described in Section 4.

Two further properties of TMR sensors matter for this project. First, the sensing signal comes from the change in junction resistance between the parallel and antiparallel magnetisation states, usually expressed as the TMR ratio $(R_{\mathrm{AP}} - R_{\mathrm{P}})/R_{\mathrm{P}}$ [5]; in a packaged bridge sensor this appears as a sensitivity in mV/V/mT. Second, the response depends on the bias voltage across the junctions: Wiśniowski et al. found that raising the bias improves sensitivity and field detection but worsens nonlinearity and hysteresis [9], so the supply voltage is a design choice rather than a fixed property of the part. Chip-scale magnetoresistive magnetometers already have CubeSat flight heritage. The AMR-based MAGIC instrument flew on CINEMA, where an attitude-independent calibration against the International Geomagnetic Reference Field gave an absolute accuracy within 2% [10], and on RadCube, where both a boom-mounted and an in-board sensor met ESA space-weather measurement requirements [4]. These missions show that low-cost chip sensors can be calibrated in orbit, but they used AMR rather than TMR sensors.

2.2 Radiation Tolerance of TMR Sensors

Because TMR is comparatively new to the space sector, published radiation-tolerance data is limited, though what exists is promising. Alves' assessment of magnetoresistive sensors for space applications supports the conclusion that TMR devices are radiation-hard [8]. Published testing of TMR magnetic sensors based on MTJ structures found their performance unchanged under bias voltages of $+5$, 0, and -5 V after a Total Ionising Dose (TID) of 5 Mrad(Si) using γ-ray and X-ray sources [11], and MTJs intended for a CubeSat magnetometer have been tested to 100 krad [12]. Ren et al. exposed MgO-barrier MTJs to gamma rays up to 10 Mrad(Si) and found no change in coercivity, exchange coupling or tunnelling magnetoresistance [13], and a 2023 review of MgO-based MTJs under gamma, X-ray and other radiation concluded that they are more tolerant than silicon devices [14]. These doses are well above the 100 krad(Si) level typically used to qualify a device as radiation-hardened. The result is physically consistent with the crystalline MgO barrier at the heart of the MTJ: ionising radiation predominantly damages devices through electron-hole pairs that become trapped at defect sites, permanently shifting

threshold voltages, a mechanism that primarily affects the *amorphous* SiO_2 gate oxide used in conventional CMOS devices [15], and for which MgO's ordered crystal lattice offers far fewer trap sites.

Commercial magnetometers are also routinely radiation-tested before use in space. Regoli et al. irradiated nine PNI RM3100 magnetometers, a COTS magneto-inductive sensor, to 300 krad(Si) and found that most units tolerated the full dose [16], and a commercial CubeSat magnetometer is specified at a component-level TID of 10 krad [17]. Testing several units of a low-cost COTS part is therefore an accepted approach. The tolerance of TMR junctions, however, has mostly been shown for the sensing element itself rather than for a packaged COTS part working in a circuit. This literature evidence motivated the TID testing described in Section 9.

2.3 Strong Magnetic Fields on CubeSats and Offset Shift in TMR Sensors

CubeSat magnetometers sit close to magnetic sources, chiefly the magnetorquers used for attitude control, and are therefore usually mounted as far from the spacecraft as possible, ideally on a boom [17]. A typical CubeSat magnetorquer board has a nominal magnetic moment of about $0.2\,\mathrm{A\,m^2}$ and can sit inside the electronics stack [18]. The on-axis field of a current loop of radius R carrying current I is $B = \mu_0 I R^2 / \left(2(r^2+R^2)^{3/2}\right)$ [19]. Far from the loop ($r \gg R$) this becomes $B = \mu_0 m/(2\pi r^3)$, where $m = I\pi R^2$ is the magnetic moment. Applied to the magnetorquer, this gives 5 mT at 2 cm, 1.5 mT at 3 cm and 0.3 mT at 5 cm. These are roughly 5 to 100 times Earth's field of 25 to 65 µT [20], but they are within the CT100's $\pm$50 mT operating range [21]. (The point-dipole estimate overstates the field at distances comparable to the coil size, so these values are an order-of-magnitude guide.) A correctly operating magnetorquer would therefore not normally saturate a TMR sensor, although it would swamp the geomagnetic signal unless the readings are timed around it. Saturation needs a much stronger or closer source, such as a permanent magnet near the sensor, a magnetorquer within about a centimetre, or magnetising equipment during ground handling and test. It is best treated as an off-nominal event, and the questions that matter are whether the sensor survives it and what it leaves behind.

The manufacturer states that its XtremeSense TMR sensors survive any magnetic field amplitude, but notes that after a strong field the offset can shift, that this shift is stable and irreversible, and that, because such events are rare, an offset recalibration is typically recommended [22]. This confirms that the effect is expected, but it gives no magnitude for a specific part and does not consider a platform on which the sensor's signal is only a few tens of microtesla, where an offset of even one millitesla would dominate. The datasheet quotes a hysteresis error of at most 0.05% for fields within $\pm$20 mT [21], but this describes behaviour inside the operating range and says nothing about what remains after the sensor has been driven beyond it. Because in-flight calibration can recover a magnetometer's offset [10], the question that matters for a CubeSat is not whether such a shift exists but how large it is and whether the rest of the response survives.

2.4 Research Gap

Taken together, the literature establishes four things. TMR is among the most sensitive chip-scale magnetoresistive technologies suitable for CubeSat-class ADCS/EPS use (Section 2.1). TMR junctions have been shown to tolerate radiation at the element level, and COTS magnetometers are routinely radiation-tested (Section 2.2). Strong fields are an off-nominal hazard for a CubeSat magnetometer, and the manufacturer acknowledges a lasting offset shift after

one (Section 2.3). Yet, in the sources found for this review, no study reports device-level data for a specific low-cost packaged COTS TMR sensor working inside a real circuit that covers (a) how quickly it recovers from saturation, in terms an on-board computer designer can use, (b) how large the lasting offset shift is and whether its gradient and linearity survive, and (c) its behaviour after irradiation, using a test method that can be reproduced with ordinary bench equipment. To the best of the author's knowledge no published study reports these for the CT100. This gap motivates the research question and objectives in Section 3.

3 Research Question, Aims, and Objectives

Research question. Is a low-cost, commercial-off-the-shelf TMR sensor (the CT100) operationally viable for CubeSat ADCS/EPS use, specifically with respect to (a) its behaviour when driven into saturation by an unexpectedly strong magnetic field and then recovered, including any lasting change to its output, and (b) any change in its sensing performance following Total Ionising Dose radiation exposure representative of the CubeSat orbital environment?

This question follows directly from the gap identified in Section 2.4: the literature supports TMR's theoretical suitability and likely radiation tolerance, and the manufacturer acknowledges an offset shift after strong fields, but it does not provide device-level evidence for a specific, inexpensive COTS part of the kind a resource-constrained CubeSat programme would actually select.

Aims. This project aims to generate that device-level evidence for the CT100, and, in doing so, to build and validate a low-cost, automatable test methodology that could be reused for other candidate COTS TMR devices without requiring specialised space-qualification equipment.

Objectives. The research question and aims above translate into the following concrete objectives, which structure the remainder of this report:

(i) characterise the TMR sensor itself – its operating principle and its saturation ceiling explored in Section 4;

(ii) design and build hardware capable of generating a known, controllable test field and logging the sensor's response to it automatically, without relying on specialised or costly space-test equipment, this can be found in Sections 5–7;

(iii) quantify the sensor's sensitivity and saturation-recovery time, and the accuracy of the field model used to drive it, establishing a non-irradiated baseline cross-validated across instruments evaluated in Sections 8 and 8.4;

(iv) measure, with a controlled before-and-after test on a fresh sensor, how saturation changes the sensor's output offset, gradient and linearity (Section 11);

(v) evaluate whether low-cost ferrite shielding materially changes the measured response (Section 10); and

(vi) assess the sensor's tolerance to Total Ionising Dose radiation representative of the CubeSat orbital environment, with and without magnetic shielding, against the same validated methodology (Section 9).

3.1 Hypotheses

The objectives above are tested through six hypotheses. Each one is stated with the criterion used to judge it, so that the results can support or reject it, and each is linked to the section where it is tested.

H1: Saturation ceiling. The output saturates at $V_{\text{out}} = \pm V_{\text{DD}}/3$, as derived from the bridge topology in Section 4.3, and the positive and negative ceilings are equal in magnitude. It is supported if every measured ceiling lies within a few percent of $V_{\text{DD}}/3$, allowing for the tolerance of the element resistances.

H2: Field model and sensitivity. The gradient of V_{out} against trace current matches the gradient predicted from the field model and the datasheet sensitivity. It is supported if the measured gradient corresponds to a sensitivity inside the datasheet range of 3.8 to 5.2 $\text{mV V}^{-1}\,\text{mT}^{-1}$ (explored in Section 8.3).

H3: Saturation recovery. After saturation the output returns to its baseline within one sample interval of a typical CubeSat magnetometer read at 10 Hz (100 ms), so no samples need to be discarded. It is supported if the worst-case recovery time, including a safety margin, is shorter than this interval (Sections 6.3 and 8.2).

H4: No lasting effect of saturation. Driving the sensor into saturation leaves no lasting change. It is supported if, on a sensor with no magnetic history, the zero-field baseline after saturation is within one ADC step of the baseline before saturation and the gradient is unchanged within session-to-session variation (Section 11).

H5: Radiation tolerance. A total ionising dose of 40.76 krad(Si) causes no measurable permanent change, as the literature on MgO-barrier MTJs predicts (Section 2.2). It is supported if the saturation ceiling, recovery time and gradient of irradiated sensors differ from non-irradiated sensors by less than the variation between non-irradiated units and sessions (Section 9).

H6: Shielding. Enclosing the sensor in ferrite shielding measurably changes its response by reducing the influence of the ambient field. It is supported if the shielded response differs from the unshielded response by more than the session-to-session variation (Section 10).

4 The CT100 Sensor

4.1 Device Overview

The CT100LW-FS6 Tunnel Magnetoresistance (TMR) sensor was selected primarily due to its thermal stability, differential output, small footprint, and cost. The device is rated for an operating temperature range of $-40\,^{\circ}\text{C}$ to $150\,^{\circ}\text{C}$ [21], making it highly suitable for thermal cycling tests relevant to my partner Daniels Tieu's work. Also, its differential output architecture provided high Common-Mode Rejection Ratio (CMRR), mitigating common-mode noise across its magnetic dynamic range of ± 50 mT with a low linearity error of $\pm 0.5\%$ (between -20 mT and $+20$ mT) [21].

Integrated TMR sensor ICs were explored but ultimately were decided against as raw signal behavior of the sensor was desired, discrete analog sensor enables direct characterization and

isolated testing of the TMR elements itself and provided additional flexibility for system integration if needed. Furthermore, its SOT-23-6 package minimizes both PCB footprint and mass, aligning with Size, Weight, and Power (SWaP) constraints typical of CubeSat applications. Sourced at a unit cost of $0.92 via DigiKey (as of April 2026) [23], it provided a highly economical solution for our testing.

The CT100LW-FS6 supports a supply voltage range of 1.0 V to 5.5 V [21]. Because the sensor's differential output sensitivity scales proportionally with supply voltage ($S \propto V_{DD}$), operating at a higher V_{DD} increases the full-scale differential output voltage swing per unit magnetic field (mV/T). This larger voltage swing enhances the signal-to-noise ratio (SNR). However, a supply voltage of $V_{DD} = 3.30\,\text{V}$ was selected to match the 3.30 V operating voltage of a STM32 microcontroller. This eliminates the need for additional voltage regulators or signal conditioning level-shifters, simplifying the power electronics if system integration was required.

The datasheet also gives the electrical characteristics used later in this report: a bridge resistance of 20 to 40 kΩ, a typical power consumption of 0.30 mW at $V_{DD} = 3.0\,\text{V}$, a zero-field offset of at most $\pm 5\,\text{mV}\,\text{V}^{-1}$, a sensitivity of 3.8 to $5.2\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$ (typically 4.5), a hysteresis error of at most 0.05% within ±20 mT, and an output noise of $700\,\text{nV}/\sqrt{\text{Hz}}$ at 10 Hz [21].

4.2 Physical Operating Principle

The CT100LW-FS6, like other TMR devices is built around magnetic tunnel junctions (MTJs): a pinned ferromagnetic reference layer, an ultra-thin (~2 nm) insulating crystalline magnesium oxide (MgO) layer and a magnetically free ferromagnetic layer [6, 7].

Conduction through the junction is done under the quantum tunnelling property, in which electrons have a nonzero probability of passing through the MgO barrier. Figure 1 denotes the MTJs equivalent circuit model.

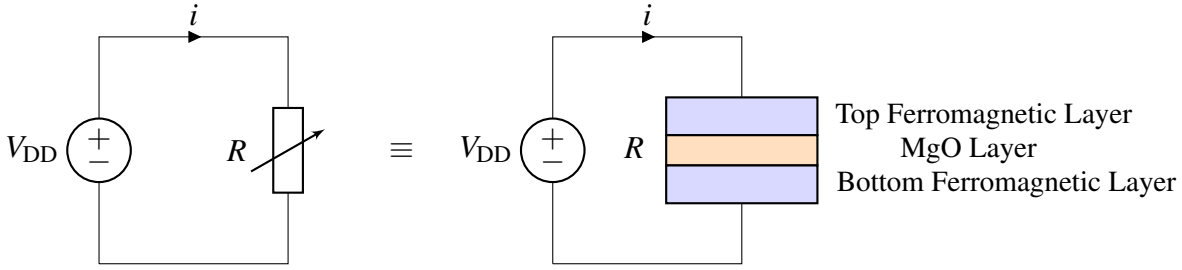


Figure 1: Equivalent electrical circuit model equating a standard variable resistor (left) to the physical three-layer Magnetic Tunnel Junction stack structure (right).

The basic operation of the MTJ comes down to how easily electrons can pass through the MgO barrier placed between two ferromagnetic layers. The bottom layer serves as a fixed reference with a locked magnetic direction, while the top free layer can rotate its magnetic orientation 360° to align an external magnetic field.

When an external field aligns the free layer's direction (0°) with the bottom reference layer, electrons hop across the insulating barrier much more easily. Higher electron movement per unit of time represents a greater amount of electrical charge ΔQ passing through the barrier over time Δt. By definition, current is the rate of charge flow per unit of time:

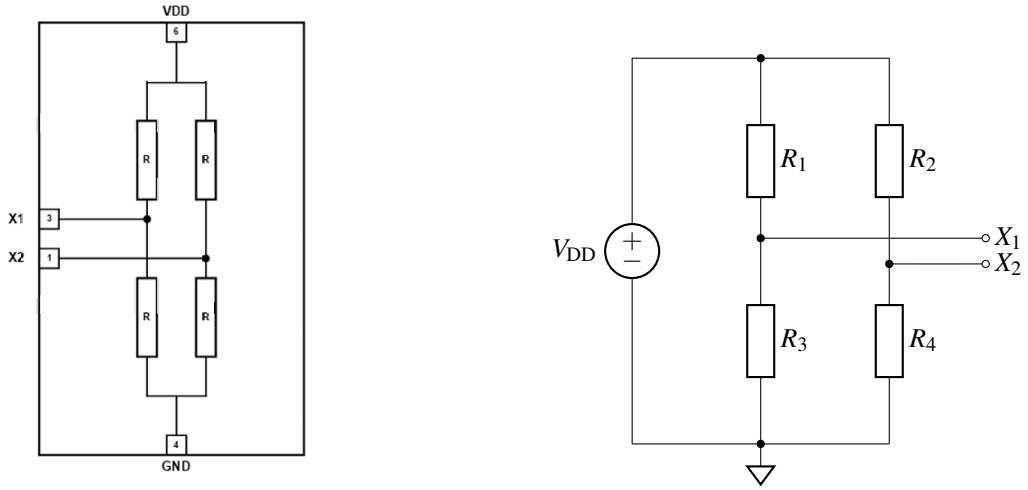
$$I = \frac{\Delta Q}{\Delta t} \tag{1}$$

Therefore, this increased charge movement results in a higher current flow I for a constant supply voltage V_{DD}, putting the sensor into its parallel, lowest resistance state $R = \frac{V_{DD}}{I}$. Conversely, when the field lines points in the opposite direction (180°), fewer electrons can cross, restricting current flow and increasing resistance. If the external magnetic field is applied orthogonal to the fixed reference direction, the sensor cannot detect it, making the element effectively blind to perpendicular fields. Hence, a TMR element can be modelled as a variable resister where it's resistance varies dependant on the polarity and strength of an external magnetic field.

fix this othongal thing just means we don't know what axis the field is more towards parlell or anti parell kinda thing

4.3 Bridge Topology, Output Convention, and Saturation Ceiling

The CT100LW-FS6 (I will refer to the device as the CT100) arranges four such TMR elements in a passive Wheatstone bridge configuration as seen in Figure 2.



(a) CT100 internal Wheatstone-bridge.

(b) Equivalent bridge schematic.

Figure 2: Internal CT100 IC pin configuration alongside its labeled equivalent electrical schematic.

To predict the maximum differential output voltage prior to saturation, the CT100's internal TMR Wheatstone bridge (Figure 2a) was evaluated theoretically. An equivalent schematic with explicitly labeled internal elements is provided in Figure 2b to aid in following the derivation below. According to the device datasheet [21], the per-element resistance ranges from $R_{\min} \approx 20\,\text{k}\Omega$ where the element experiences parallel magnetic alignment, lowest-resistance state) to $R_{\max} \approx 40\,\text{k}\Omega$ (antiparallel magnetic alignment, highest-resistance state).

As a result of this Wheatstone-bridge configuration, the largest change in $|V_{\text{out}}|$ is $\frac{V_{DD}}{3}$:

$$|\Delta V_{\text{out,max}}| = \frac{V_{DD}}{3} \tag{2}$$

This maximum output swing is bounded by each of the internal TMR element resistance range ($R_{\min} \approx 20\,\text{k}\Omega$ to $R_{\max} \approx 40\,\text{k}\Omega$).

The voltage potentials at the two central output nodes, X_1 and X_2, as seen in Figure 2b are

determined by the independent voltage divider ratios along the bridge:

$$X_1 = V_{\mathrm{DD}} \cdot \left(\frac{R_3}{R_1 + R_3} \right), \qquad X_2 = V_{\mathrm{DD}} \cdot \left(\frac{R_4}{R_2 + R_4} \right) \tag{3}$$

The general differential output voltage V_{out} across nodes X_1 and X_2 is expressed as:

$$V_{\mathrm{out}} = X_1 - X_2 = V_{\mathrm{DD}} \left(\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right) \tag{4}$$

At $B = 0\,\mathrm{T}$, all four TMR elements exhibit nominal, equal resistance ($R_1 = R_2 = R_3 = R_4$), yielding balanced potentials $X_1 = X_2 = \frac{V_{\mathrm{DD}}}{2}$ and $V_{\mathrm{out}} = 0\,\mathrm{V}$.

A key operational characteristic of the CT100 is that an external magnetic field applied along its sensitive axis drives opposing element pairs in opposite resistance directions. As one pair (R_1, R_4) decreases toward R_{min}, the complementary pair (R_2, R_3) increases toward R_{max} or vice versa. This response is governed by the orientation of the pinned ferromagnetic reference layer within each TMR element: R_1 and R_4 share an identical reference layer alignment, whereas R_2 and R_3 are pinned in the opposite direction.

At full magnetic saturation, the opposing resistor pairs settle at either R_{min} and R_{max}. For this derivation, an external magnetic field polarity is assumed such that:

$$R_1 = R_4 = R_{\mathrm{min}}, \quad \text{and} \quad R_2 = R_3 = R_{\mathrm{max}} \tag{5}$$

Substituting these saturation values into Equation 3 gives the node potentials at full saturation:

$$X_{1,\mathrm{sat}} = V_{\mathrm{DD}} \cdot \left(\frac{R_{\mathrm{max}}}{R_{\mathrm{min}} + R_{\mathrm{max}}} \right), \qquad X_{2,\mathrm{sat}} = V_{\mathrm{DD}} \cdot \left(\frac{R_{\mathrm{min}}}{R_{\mathrm{max}} + R_{\mathrm{min}}} \right) \tag{6}$$

Subtracting $X_{2,\mathrm{sat}}$ from $X_{1,\mathrm{sat}}$ yields the maximum theoretical differential output voltage $V_{\mathrm{out,sat}}$:

$$V_{\mathrm{out,sat}} = X_{1,\mathrm{sat}} - X_{2,\mathrm{sat}} = V_{\mathrm{DD}} \left(\frac{R_{\mathrm{max}} - R_{\mathrm{min}}}{R_{\mathrm{max}} + R_{\mathrm{min}}} \right) \tag{7}$$

Substituting the CT100 datasheet resistance values [21] ($R_{\mathrm{min}} = 20\,\mathrm{k\Omega}$ and $R_{\mathrm{max}} = 40\,\mathrm{k\Omega}$):

$$V_{\mathrm{out,sat}} = V_{\mathrm{DD}} \left(\frac{40\,\mathrm{k\Omega} - 20\,\mathrm{k\Omega}}{40\,\mathrm{k\Omega} + 20\,\mathrm{k\Omega}} \right) = V_{\mathrm{DD}} \left(\frac{20}{60} \right) = \frac{V_{\mathrm{DD}}}{3} \tag{8}$$

This derivation proves that the maximum differential output voltage is bound to $|\Delta V_{\mathrm{out,max}}| = \frac{V_{\mathrm{DD}}}{3}$. For a supply voltage of $V_{\mathrm{DD}} = 3.30\,\mathrm{V}$, the theoretical saturation output $|V_{\mathrm{out,sat}}|$ is $1.10\,\mathrm{V}$. This is experimentally validated in Section 6.3.

The direction of the applied field along the sensitive axis decides which resistor pair moves to R_{min}, so the bridge can saturate at either polarity. The two limits, $V_{\mathrm{out}} = +V_{\mathrm{DD}}/3$ and $V_{\mathrm{out}} = -V_{\mathrm{DD}}/3$, are called the **positive ceiling** and the **negative ceiling** respectively. The negative ceiling follows from the derivation above by swapping R_{min} and R_{max}. In this report, *ceiling* means the saturation limit of V_{out}, and its sign states which of the two was reached.

Furthermore, from this point forward in this report, the sensor's output voltage is defined as $V_{\mathbf{out}} = X_1 - X_2$, which was applied across all DMM, oscilloscope, and MCU measurement in this work.

5 Available Instrumentation

Characterization and bench-top tests were conducted using the following laboratory equipment to ensure power delivery and take measurements.

- **Digital Oscilloscope:** Rohde & Schwarz RTB2002 (2.5 GS/s sampling rate, 10-bit A/D converter) for transient analysis, step-response monitoring across nodes X_1 and X_2.
- **Digital Multimeter (DMM):** Keithley 2110 $5\frac{1}{2}$ Digit Dual-Display Multimeter for DC Voltage and Current logging and cross-validating MCU reading accuracy.
- **DC Power Supply Unit (PSU):** Topward dual-tracking linear DC power supply capable of supplying up to 6.0 A continuously, providing supply voltage and current excitation through the PCB test trace.

6 Standalone Sensor Board

Before adding any electronics, the bare sensor was characterised on its own. This gives a reference for how the CT100 itself behaves, so that any later change in behaviour can be attributed to the circuit, to radiation or to saturation rather than to the sensor. The results answer H1 (the saturation ceiling) and give the first measurement for H3 (recovery time).

6.1 PCB Design

To quantify the CT100 saturation limits and verfiy the derivations conducted in Section 4.3, a standalone sensor board was designed in Altium Designer, the assembled boards were populated with just the CT100 sensor and a single 0.1 µF decoupling capacitor.

The board is driven directly from a laboratory PSU, with the differential outputs X_1, X_2 measured by a digital oscilloscope and DMM. Signal amplification was deliberately excluded from the design. Although raw bridge differential voltages can be relatively small, avoiding amplifier stages eliminates amplifier constraints such as slew rate and bandwidth obscuring the true recovery time of the TMR bridge itself. Probing the unconditioned Wheatstone output directly establishes a true baseline of the sensor's native sensitivity and saturation recovery. This design was also used in Daniel Tieu's work on Thermal Cycling. The PCB can be seen in Figure 3.

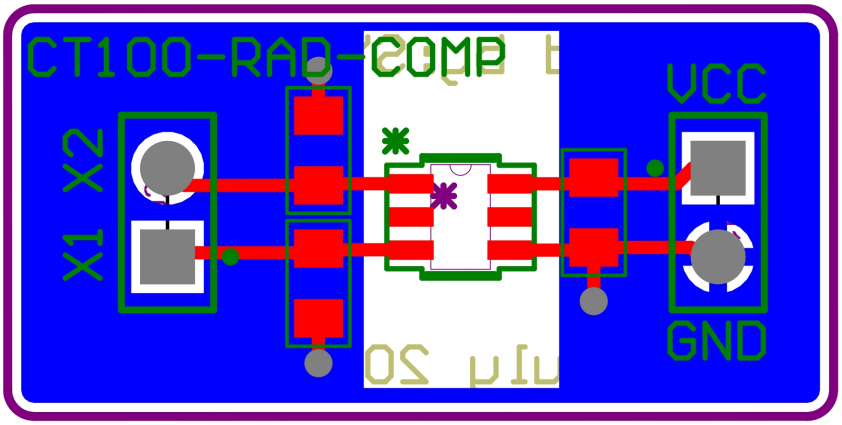


Figure 3: Standalone Sensor PCB

6.2 Saturation Testing Methodology

For this board, magnetic saturation was induced using an external LM-1618 Neodymium magnet. This manual method of bringing the magnet toward and away from the sensor was the simplest available stimulus for an initial characterization, though it is not inherently repeatable or automatable.

To reliably drive the device into saturation and characterize the sensor's transient recovery and hysteresis under worst case situations, the device was intentionally driven far beyond its standard operating limits. The CT100 sensor has an operating magnetic field range of $\pm 50\,\mathrm{mT}$. However, its absolute maximum threshold where device reliability may be affected is specified at $\pm 200\,\mathrm{mT}$ [21].

The magnet used was an LM-1618 neodymium disc, 25 mm in diameter and 5 mm thick, with a remanence B_r of 1180 to 1220 mT [24]. The remanence is a property of the material, not the field outside the magnet. On the axis of a disc magnet of radius R and thickness L, treating the magnet as an ideal solenoid [25, 26], the field at distance z from its face is

$$B(z) = \frac{B_r}{2}\left[\frac{L+z}{\sqrt{R^2+(L+z)^2}} - \frac{z}{\sqrt{R^2+z^2}}\right], \tag{9}$$

which gives about 223 mT at the face, 152 mT at 5 mm and 86 mT at 10 mm. Within about a centimetre the field is therefore well beyond the CT100's $\pm 50\,\mathrm{mT}$ operating range, which guarantees saturation, while staying close to or below its 200 mT absolute-maximum rating [21], since the package itself keeps the sensing element away from the magnet face.

During testing, the differential output voltage was monitored using an oscilloscope to confirm that it hit the saturation limit of $|\Delta V_{\mathrm{out,max}}| \approx V_{\mathrm{CC}}/3$ which we derived in Section 4.3. Once the output voltage was confirmed to be at this saturation ceiling, the magnet was rapidly removed. The oscilloscope then captured the transient recovery, recording the time required for the output to settle back to a steady-state baseline under no external magnetic field excitation.

Every saturation result in this report is presented in the same table format, with the same definitions:

- **Sample:** the unit tested. Samples 1 to 4 are sensor-only boards (1 and 2 non-irradiated, 3 and 4 irradiated). Boards A, B and C are integrated boards (A non-irradiated, B irradiated, C a fresh sensor used for the hysteresis test in Section 11).
- **Ceiling (V):** the output voltage V_{out} reached at saturation. Its sign shows which ceiling was reached, so each sample has one row per ceiling.
- **Baseline (mV):** the output once it has settled after the field is removed.
- **Swing (V):** the size of the drop from the ceiling to the baseline, $|\text{ceiling} - \text{baseline}|$.
- **Recovery (ms):** the time from the start of the fall until the output has settled at its baseline. On the oscilloscope this is read with cursors; for the MCU (SWV) traces it is the time until the output stays within 5 mV of its final baseline. Five millivolts is also about the oscilloscope's vertical resolution at 500 mV/div, so the two definitions of "settled" are comparable.

6.3 Sensor-Only Saturation and Recovery Behaviour

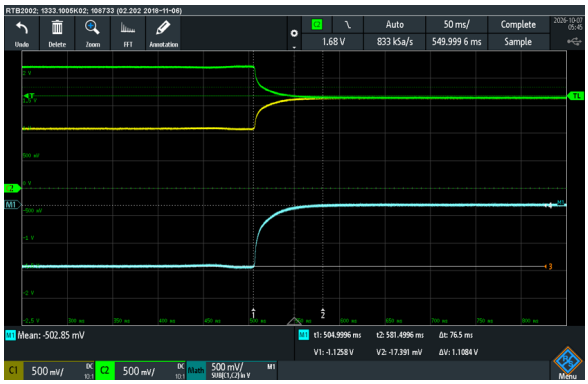
Two non-irradiated sensor-only samples (Samples 1 and 2) were each driven to both ceilings. Testing both directions checks a prediction of the bridge model: the ceilings should be symmetric about zero, $\pm V_{\text{DD}}/3$ (Section 4.3), because the same spin-tunnelling limit applies to either polarity. The extracted values are listed in Table 1, and the oscilloscope captures they come from are shown in Figure 4.

The CT100 is sensitive to the field component along one in-plane axis of its package. The datasheet defines a positive field as one directed into the pin 1/pin 6 side of the package, which makes $X_1 > X_2$ [21]. Which ceiling is reached therefore depends only on which pole of the magnet faces that side of the package: one pole drives V_{out} to the positive ceiling and the opposite pole to the negative ceiling.

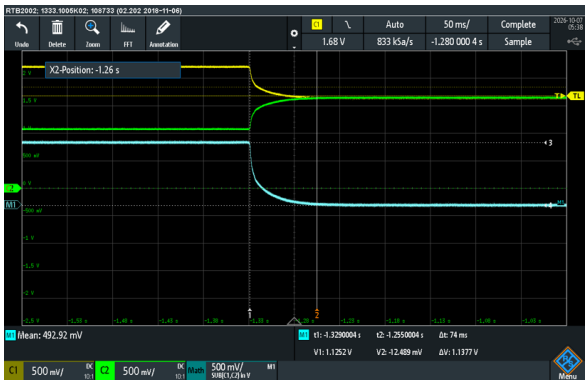
Table 1: Saturation recovery results, non-irradiated sensor-only samples.

Sample	Ceiling (V)	Baseline (mV)	Swing (V)	Recovery (ms)
1	−1.126	−17.4	1.108	76.5
1	+1.125	−12.5	1.138	74.0
2	−1.145	−17.4	1.128	76.0
2	+1.125	−12.5	1.138	67.5

To visually document the transient response across both polarities of saturation recovary, Figure 4 presents the oscilloscope captures for Samples 1 and 2. The traces highlight the response as the differential output transitions from the saturated ceiling back to the steady-state baseline once the Neodymium magnet is removed. The blue waveform (M_1) represents the differential output voltage ($V_{\text{out}} = X_1 - X_2$), while the yellow ($C_1$) and green ($C_2$) waveforms represent the individual X_1 and X_2 bridge terminal voltages, respectively.



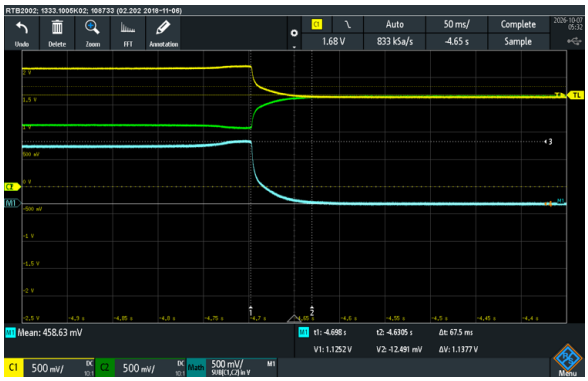
(a) Sample 1: Negative Saturation Recovery



(b) Sample 1: Positive Saturation Recovery



(c) Sample 2: Negative Saturation Recovery



(d) Sample 2: Positive Saturation Recovery

Figure 4: Oscilloscope traces capturing the saturation ceiling and subsequent transient recovery for the non-irradiated sensors.

Across the four captures the mean recovery time is 73.5 ms, with a maximum of 76.5 ms. Because the magnet was withdrawn by hand, these times include the effect of the withdrawal speed. What they mean for a CubeSat magnetometer, including the guard window that flight software would need after a saturating event, is assessed in Section 12.3 once all samples have been presented.

7 Integrated Board: Field Excitation and Design

A hand-held magnet cannot produce a known field, so it cannot measure sensitivity, and it cannot be automated. The integrated board solves both problems: a current-carrying trace produces a field that can be calculated from the measured current, and the microcontroller logs the sensor output without an operator. This board is the low-cost, reusable test method named in the aims, and it is used for every test of sensitivity (H2), lasting change (H4), radiation (H5) and shielding (H6).

7.1 Integrated Board Design

To accurately characterize the sensor's sensitivity without relying on a handheld neodymium magnet, a repeatable and quantifiable test field was required.

A current-carrying PCB trace routed directly beneath the sensor was selected. This DC current induces a magnetic field around the trace that acts upon the TMR sensor to generate a measurable voltage output. It's generated magnetic field is calculated analytically and simulated in Ansys Maxwell, serving as a theoretical benchmark for expected sensor output against practical results. The trace current can be swept up to the laboratory PSU's 6.0 A limit.

While the upper limit of the PSU capability 6.0 A does not generate a strong enough magnetic field to drive the CT100 into saturation, it provides a linear sweep to determine the sensor's sensitivity. Establishing this sensitivity benchmark is critical as later in the characterization, this baseline allows us to measure any resulting hysteresis or degradation in linearity under saturation and/or radiation.

The system architecture of the integrated board revolves around three core components:

- **Microcontroller (MCU):** An STM32G431KBT6 [27] was selected for it's high-performance. The CT100 differential output pair feeds the MCU's internal differential ADC channels directly, sampled at 1.0 kHz. The MCU features a 12-bit ADC and auto-calibration routines were run to minimise discrepancies between ADC's on MCU on different boards.
- **Current Monitor (INA219) [28]:** Because the PSU's analog display was not trusted as precise means of determining current, a dedicated INA current-sense with a Kelvin-connected 5.0 mΩ shunt resistor. This was used to back-calculate the magnetic field generated by the trace current acting on the TMR sensor.
- **Low-Dropout (LDO) regulator:** The board is powered via a 5.0 V input rail which is fed into into a LDO regulator to provide a clean 3.30 V supply to both the sensor and MCU. This LDO also provides a high Power Supply Rejection Ratio (PSRR) and sets the subsequent resolution of the TMR sensor.

As shown in Figure 5, the current carrying trace is routed on the bottom layer indicated by the blue polygon with a 2512 shunt resister in series. Figure 6's schematic denotes the positive current ($+i$) sign convention adopted for all subsequent field calculations and experimental sweeps. The Altium schematic for the integrated design can be found in the Appendix A.

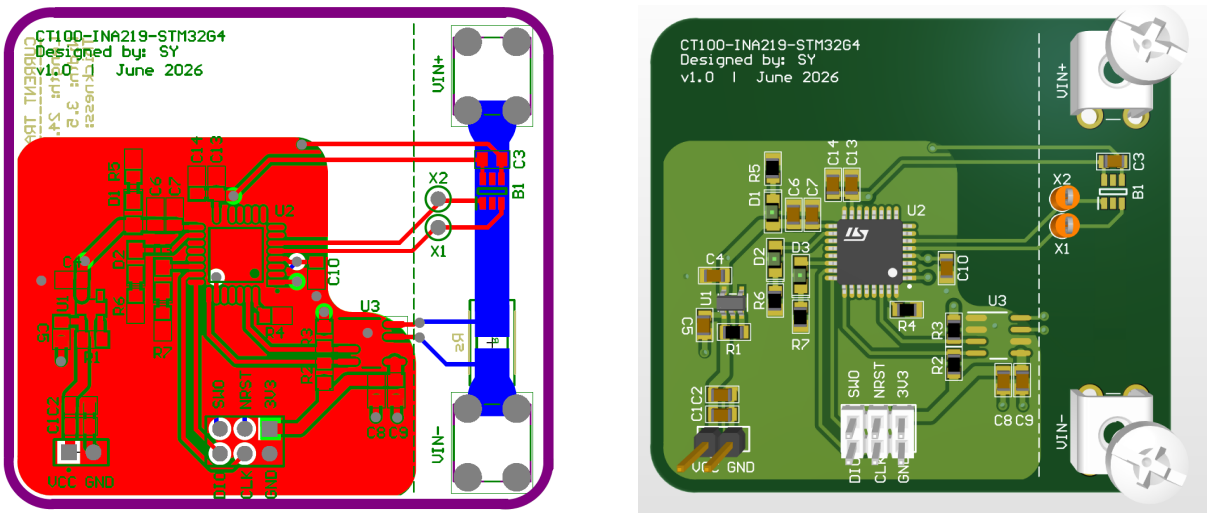


Figure 5: Integrated board layout (left) and rendered assembly view (right).

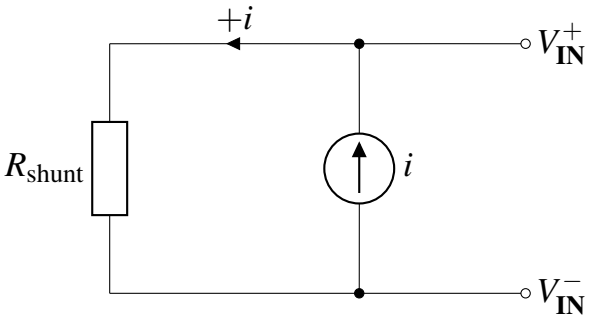


Figure 6: Field excitation circuit schematic and positive trace current convention ($+i$).

Note that while Figure 6's models the excitation as an ideal current source i, the physical setup employs a benchtop PSU operated in Constant Current (CC) mode with the current limit determining the current through R_{shunt} and the trace.

7.2 PCB Trace Design

The current-carrying trace was routed on a four-layer PCB, with layer thicknesses and dielectric parameters detailed in Table 2. The trace was routed such that current flow produces a field acting on the sensor's most senstive axis, where no copper planes placed near the TMR device and the current trace.

No copper pour was placed on any layer directly beneath the sensor, so that no plane or return currents flow close to the sensing element and the only board-generated field at the die is that of the excitation trace. Copper is non-magnetic, so the concern is current flowing in nearby copper rather than the copper itself distorting a static field.

Table 2: Four-layer PCB stack-up, from the Altium Layer Stack Manager.

Layer	Material	Type	Thickness (mm)	D_k
Top solder mask	SM-001	Solder mask	0.0254	4.0
Top layer	CF-004, 1 oz	Signal	0.0350	—
Dielectric 4	PP-017	Prepreg	0.1295	4.3
Dielectric 2	PP-017	Prepreg	0.1295	4.3
Layer 2 (GND)	CF-001, 1/4 oz	Signal	0.0090	—
Dielectric 1	FR-4	Core	0.3200	4.8
Layer 3 (PWR)	CF-001, 1/4 oz	Signal	0.0090	—
Dielectric 3	PP-017	Prepreg	0.1295	4.3
Dielectric 5	PP-017	Prepreg	0.1295	4.3
Bottom layer (trace)	CF-004, 1 oz	Signal	0.0350	—
Bottom solder mask	SM-001	Solder mask	0.0254	4.0
Total			0.9770	

The board is 0.977 mm thick in total: 0.088 mm of copper and 0.838 mm of dielectric, made up of four 0.130 mm prepreg layers and a 0.320 mm FR-4 core. The excitation trace is on the bottom layer, with a width $w = 3.5$ mm, sized to carry the PSU's 6.0 A limit.

7.3 Analytical Field Model

The horizontal magnetic field at the sensing element, a height d above the centre of the trace of width $w = 3.5\,\mathrm{mm}$, is modelled as the field of a thin flat strip conductor:

$$B_{\mathrm{planar}} = \mu_0 \left[\frac{I}{\pi w} \arctan\left(\frac{w}{2d}\right) \right]. \tag{10}$$

The trace is treated as a thin flat strip of width w carrying the current I uniformly, so it can be split into long straight filaments of width $\mathrm{d}x$, each carrying $\mathrm{d}I = (I/w)\,\mathrm{d}x$. A long straight filament produces a field $\mu_0\,\mathrm{d}I/(2\pi r)$ at distance r [19]. At a height d above the centre of the strip, the vertical components from filaments on either side cancel by symmetry, and the horizontal components add, so

$$B = \int_{-w/2}^{w/2} \frac{\mu_0 I}{2\pi w} \frac{d}{x^2 + d^2}\,\mathrm{d}x = \frac{\mu_0 I}{\pi w} \arctan\left(\frac{w}{2d}\right), \tag{11}$$

which is Equation (10).

The distance d is measured from the mid-plane of the bottom-layer trace to the sensing element. From Table 2, that mid-plane lies 0.909 mm below the top copper surface: half of the 0.035 mm trace, the 0.838 mm of dielectric, the two 0.009 mm inner copper layers and the 0.035 mm top copper. The datasheet places the sensing element $0.76 \pm 0.087\,\mathrm{mm}$ above the bottom of the SOT23-6 package body, and the body stands a typical 0.10 mm (0 to 0.15 mm) above the board [21]. The sensing element is therefore $h_s \approx 0.86\,\mathrm{mm}$ above the top copper, and

$$d = 0.909\,\mathrm{mm} + h_s \approx 1.77\,\mathrm{mm}. \tag{12}$$

The solder joint under the leads adds a few tens of micrometres and is neglected. Taking the tolerances on the sensor position and the package stand-off together, d lies between 1.58 and 1.91 mm.

Normalizing Equation (10) by the trace current I yields the geometric field coupling coefficient k_B:

$$k_B = \frac{B_{\mathrm{planar}}}{I} = \frac{\mu_0}{\pi w} \arctan\left(\frac{w}{2d}\right) \approx 0.0892\,\mathrm{mT\,A^{-1}}, \tag{13}$$

with a range of 0.085 to $0.096\,\mathrm{mT\,A^{-1}}$ over the tolerance on d. At a representative excitation current of $I = 4.175\,\mathrm{A}$ this gives a field of $372\,\mu\mathrm{T}$. Combining k_B with the CT100 sensor's sensitivity S_V ($\mathrm{mV\,mT^{-1}}$) gives the expected differential voltage change per unit trace current:

$$\frac{\Delta V_{\mathrm{out}}}{\Delta I} = k_B \cdot S_V. \tag{14}$$

This analytical transfer ratio serves as the theoretical benchmark against which both the Ansys Maxwell finite-element simulations and empirical linear current sweeps are validated.

7.4 Simulated vs. Experimental Field Validation

The analytical field model was cross-checked against a finite-element simulation in Ansys Maxwell (Figure 7), which resolves the full 3D field around the trace rather than the infinite

2D strip assumed by Equation (10). The simulation gave about 500 µT at 4.175 A on its evaluation plane. In the analytical model this field occurs at a height of about 1.0 mm, closer to the trace than the datasheet sensor position of 1.77 mm, which indicates that the simulation evaluated the field on a plane closer to the trace than the sensing element; at the same height the two models agree. **[TODO: Re-run the Ansys model with the field evaluated at $d = 1.77$ mm, the datasheet sensor height, and update this paragraph and Figure 7. The analytical model predicts 372 µT there.]**

The measured output at 4.175 A (about 16 mV on the DMM) cannot be compared directly with the model, because it includes the sensor's zero-field offset of about 13 mV on this board. The like-for-like comparison is the gradient of output against current, which a fixed offset does not affect. It is made in Section 8.3, where the measured gradient agrees with the model to within a few percent.

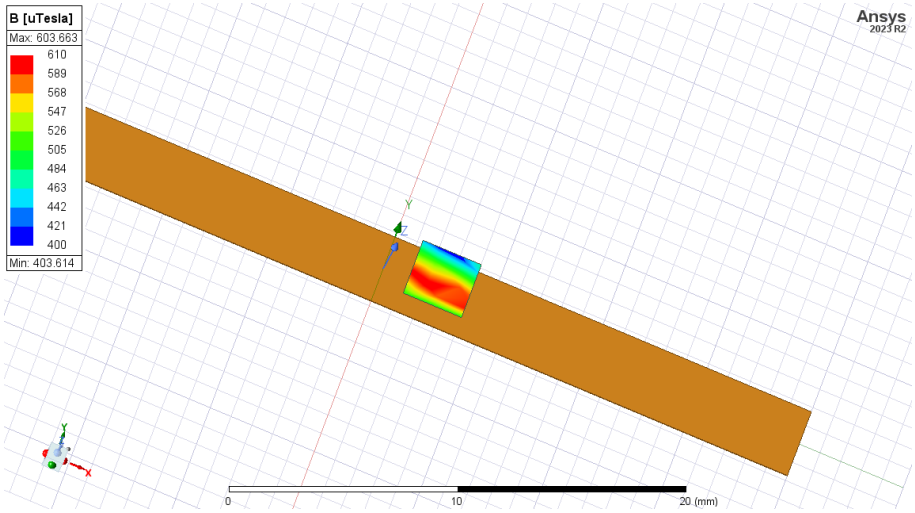


Figure 7: Ansys near-field simulation of the current trace.

7.5 Firmware

The firmware is built upon the STM32 Hardware Abstraction Layer (HAL). The MCU's ADC is configured in differential-pair mode, sampling $X_1 - X_2$ directly and converting the raw 12-bit value to volts via:

$$X_1 - X_2 = \left(\frac{\mathrm{ADC}_{\mathrm{raw}} - 2048}{2048} \right) \times V_{\mathrm{DD}} \tag{15}$$

where 2048 is the mid-scale ADC code corresponding to the zero-differential (balanced-bridge) midpoint. Hardware ADC self-calibration (`HAL_ADCEx_Calibration_Start`) is executed automatically on every boot before any conversions occur. The MCU continuously polls the INA219 over I^2C for trace current and streams both the current and voltage values out in real time over the Serial Wire Viewer (SWV) debug channel for offline analysis in MATLAB.

7.6 Design Note for Future PCB Revisions

Because the sensor bridge is nominally balanced at $V_{\mathrm{DD}}/2$ as noted in Section 4.3, the ADC channel monitoring it spends its full dynamic range on either side of that midpoint. Any mea-

surement taken near the zero-differential code inherits the ADC's own quantization and calibration uncertainty around that center point.

A future revision of the integrated board could add a small reference offset at the ADC input, so that zero field no longer sits exactly on the mid-scale code. This should reduce the error from operating at the ADC's most quantisation-sensitive point; it was not implemented or tested in this project.

8 Non-Irradiated Sensitivity and Saturation Characterisation

8.1 Linear Fitting and the Coefficient of Determination

Every current sweep in this report is summarised by a straight line,

$$V_{\text{out}} = mI + c, \tag{16}$$

where m is the gradient in mV A^{-1} and c is the offset, the output at zero trace current. The line is fitted by ordinary least squares, which chooses m and c to minimise the sum of the squared differences between the measured outputs V_i and the fitted values $\hat{V}_i = mI_i + c$.

How well the line describes the data is given by the coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (V_i - \hat{V}_i)^2}{\sum_i (V_i - \bar{V})^2}, \tag{17}$$

where $\bar{V}$ is the mean of the measured outputs. The numerator is the scatter left over after the fit and the denominator is the total spread of the data, so R^2 is the fraction of the variation in the output that the straight line explains. A value of 1 means every point lies exactly on the line; a value of 0 means the line explains none of the variation. In this report R^2 is used only as a measure of linearity. A high R^2 shows that the output is proportional to current over the sweep, but it does not show that the gradient is correct; that is checked separately against the theoretical gradient. R^2 is also unaffected by a constant offset, so a sweep with a large zero-field offset can still have R^2 close to 1.

The uncertainty of each gradient is given by its standard error,

$$s_m = \sqrt{\frac{\sum_i (V_i - \hat{V}_i)^2/(n-2)}{\sum_i (I_i - \bar{I})^2}}, \tag{18}$$

where n is the number of points in the sweep (13 for every sweep here). Gradients are reported as $m \pm s_m$. Two gradients are treated as different only if they differ by clearly more than their combined standard error, $\sqrt{s_{m,1}^2 + s_{m,2}^2}$.

8.2 Saturation Recovery on the Integrated Board

This section features evaluating th difference between the scope and intergrated board under the saturation conditions. On the integrated board, with a regulator input of $V_{\text{CC}} = 4.97\,\text{V}$ and a TMR sensor supply of $V_{\text{DD}} = 3.32\,\text{V}$.

Table 3: Saturation recovery results, non-irradiated integrated board (Board A).

Sample	Ceiling (V)	Baseline (mV)	Swing (V)	Recovery (ms)
A	−1.152	−8.1	1.144	65.2
A	+1.138	+12.9	1.125	60.2

(a) Non-irradiated, negative saturation recovery (b) Non-irradiated, positive saturation recovery

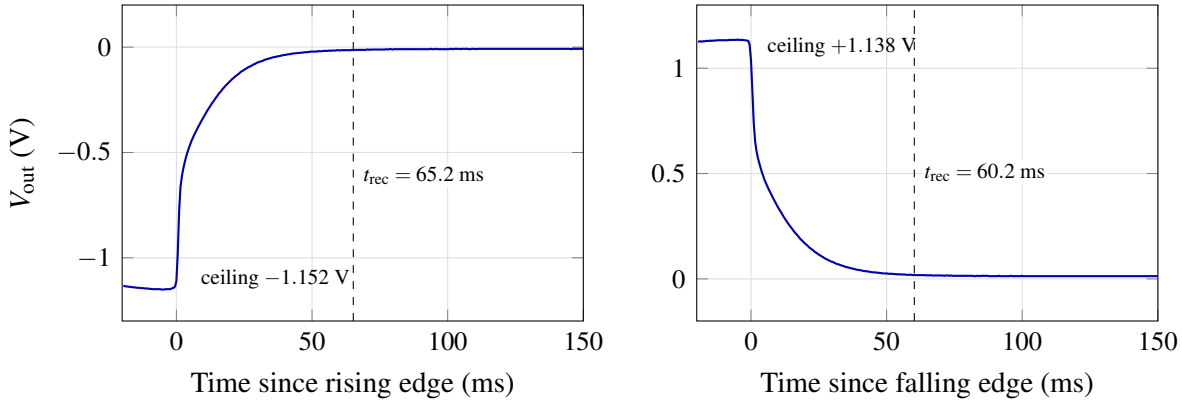


Figure 8: Non-Irradiated Saturation recovery on the integrated board after the magnet is removed, for both field directions.

Comparing the saturation behaviour of the integrated board against the standalone sensor baselines (Section 6.3) reveals highly consistent performance across both hardware configurations. In both test environments, the TMR elements exhibited symmetric saturation limits. The integrated board reached voltage ceilings of $-1.152\,\mathrm{V}$ and $1.138\,\mathrm{V}$, which closely match the $-1.145\,\mathrm{V}$ and $1.125\,\mathrm{V}$ extremes observed in the standalone samples.

The transient recovery times demonstrate tight alignment between the two configurations. The integrated board achieved full baseline restoration slightly faster of $t_{\mathrm{rec}} \approx 60.2$–$65.2\,\mathrm{ms}$ than the standalone sensors $t_{\mathrm{rec}} \approx 67.5$–$76.5\,\mathrm{ms}$. As noted in Section 6.2, the physical removal of the Neodymium magnet was performed manually.

8.3 Current Sweep: Data and Sensitivity Fit

Current-sweep data was logged via the MCU's SWV output, exported, and processed in MATLAB. Rather than relying on single instantaneous readings, each current setting was captured over a 20-second window and averaged. A single-sample readings were found to be unreliable due to the MCU ADC's quantisation behaviour where raw SWV data at a given current setting was observed hopping between a small number of discrete levels spaced at multiples of the 1.61 mV LSB typically spanning 1–2 LSBs of scatter around the true mean. Note that for this non-irradiated integrated board the saturation recovery test was performed before the current-sweep test.

The trace current was swept from -3.0 to $+3.0\,\mathrm{A}$ in approximately $0.5\,\mathrm{A}$ steps, all within a single session at $V_{\mathrm{DD}} = 3.32\,\mathrm{V}$ (Table 4).

Table 4: Non-Irradiated & No Shielding Current Sweep Data

DMM Current (mA)	INA Current (mA)	DMM (mV)	MCU (mV)	Scope (mV)
−2991.0	−2966.7	8.634	8.069	−12.207
−2498.5	−2478.2	9.860	9.571	−10.986
−2029.0	−2012.6	9.950	9.584	−10.376
−1501.0	−1487.7	11.690	11.249	−9.766
−1022.0	−1016.6	11.250	11.107	−9.308
−499.0	−498.2	12.650	12.022	−10.681
−0.7	−6.3	12.980	12.847	−8.594
499.0	487.8	14.450	13.628	−7.477
1022.0	1005.2	14.210	13.796	−5.035
1501.0	1477.2	15.697	14.790	−6.104
2029.0	2002.1	15.720	15.020	−3.516
2498.5	2466.4	16.500	16.068	−3.906
2991.0	2954.8	16.570	16.088	−5.030

A linear fit of V_{out} against trace current gives a gradient (the sensitivity to trace current) of $1.356\,\text{mV}\,\text{A}^{-1}$ on the DMM and $1.315\,\text{mV}\,\text{A}^{-1}$ on the MCU.

The CT100 datasheet [21] specifies a typical full-bridge sensitivity of $4.5\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$, with a specified range of $3.8\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$ to $5.2\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$. At the 3.32 V supply voltage used for this characterisation, the corresponding theoretical voltage sensitivity is therefore

$$S_{V,\text{theory}} = S_{\text{CT100}} V_{\text{DD}} = \left(4.5\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}\right)(3.32\,\text{V}) = 14.94\,\text{mV}\,\text{mT}^{-1}. \tag{19}$$

The analytical field model in Equation (10) gives a field coefficient of approximately $0.0892\,\text{mT}\,\text{A}^{-1}$ for the trace geometry used in the test. Combining this field response with the CT100 sensitivity gives the expected voltage change per unit trace current:

$$\frac{\Delta V_{\text{out}}}{\Delta I} = S_{V,\text{theory}} \frac{\Delta B}{\Delta I} = \left(14.94\,\text{mV}\,\text{mT}^{-1}\right)\left(0.0892\,\text{mT}\,\text{A}^{-1}\right) = 1.332\,\text{mV}\,\text{A}^{-1}. \tag{20}$$

Thus, for a given trace current, the theoretical sensor output can be written as

$$V_{\text{out,theory}} = V_0 + \left(1.332\,\text{mV}\,\text{A}^{-1}\right) I, \tag{21}$$

where V_0 is the zero-field output voltage. Figure 9 below uses a V_0 value of 12.980 mV to visually compare the voltage *slope*; it is not a theoretical prediction of the CT100 offset voltage. Figure 10 plots how close the INA device follows the DMM measurement.

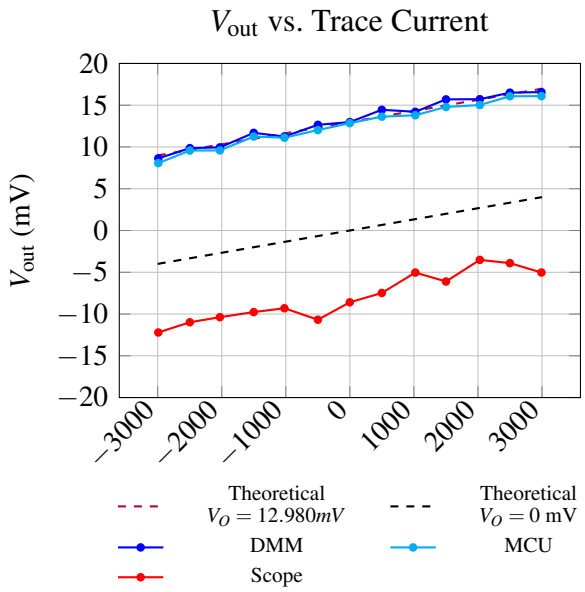


Figure 9: Voltage cross-validation during the non-irradiated, unshielded current sweep. The dashed lines shows the theoretical CT100 responses.

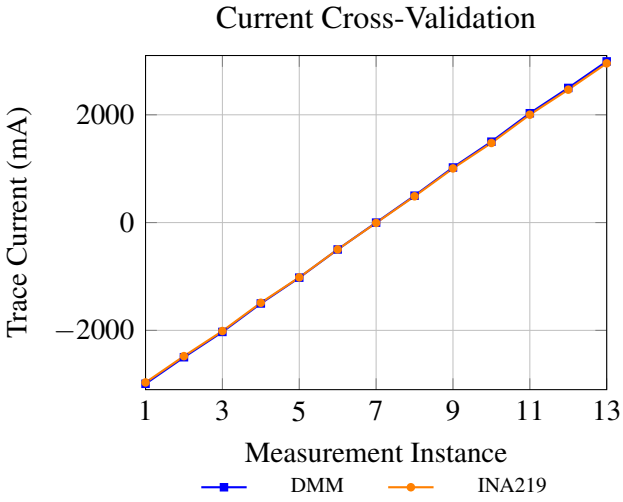


Figure 10: Current cross-validation across the 13-point sweep.

The theoretical current-to-voltage gradient at $V_{DD} = 3.32\,\text{V}$ is $1.332\,\text{mV}\,\text{A}^{-1}$ (Equation (20)). The measured DMM gradient ($1.356 \pm 0.067\,\text{mV}\,\text{A}^{-1}$) is 101.8% of this value and the MCU gradient ($1.315 \pm 0.055\,\text{mV}\,\text{A}^{-1}$) is 98.7%; both differences are smaller than the standard error of the fit. Expressed as sensitivities these are 4.58 and $4.44\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$, close to the datasheet typical value of 4.5. The tolerance on the sensor's height in its package (Section 7.3) gives a predicted gradient of 1.27 to $1.43\,\text{mV}\,\text{A}^{-1}$ (-5% to $+7\%$), so the measurement agrees with the model within both the measurement and the model uncertainty.

8.4 Instrumentation Validation

The sweep above also compares the three instruments against each other, using Figure 9 for voltage and Figure 10 for current. The DMM and MCU voltage readings closely track one another: across the 13 points the mean difference is about 0.5 mV and the largest is 0.9 mV, which is less than one ADC step (1.61 mV). The oscilloscope, in contrast, shows a different, current-dependent response. A likely cause is that its two channels are referenced to separate ground clips, so ground-return voltage drops that scale with the trace current appear differently on each channel before the channels are subtracted. This was not investigated further because the DMM and MCU already agree. The current-measurement cross-validation showed that the INA219 tracked the DMM closely across the full current sweep, with a near-unity linear relationship and a maximum difference of only 36.2 mA. The INA219 therefore provides sufficient accuracy for the current checks required in the subsequent testing. As the INA219 is already integrated into the measurement system, it will be used as the primary current measurement for future sweeps. This removes the need to repeat each current measurement using the external DMM and reduces the time required for data collection while maintaining a validated measurement reference.

The oscilloscope measurements were not consistent with the DMM and MCU voltage measurements during the DC current sweep, showing a substantially different response and negative

output values. Since the DMM and MCU agreed closely with one another, oscilloscope measurements are not used for subsequent DC voltage or sensitivity checks. The oscilloscope is reserved for transient measurements, where its time-domain capability provides information that cannot be obtained from the DMM. voltage measurements during the DC current sweep, showing a substantially different response and negative output values. Since the DMM and MCU measurements agreed closely with one another, while the oscilloscope response differed significantly. Consequently, oscilloscope measurements will not be used for subsequent DC voltage or sensitivity checks. The oscilloscope will instead be reserved for transient measurements, where its time-domain capability provides information that cannot be obtained from the DMM.

The saturation events were also found to move the zero-field baseline of the sensor. This is quantified in a dedicated before-and-after test in Section 11, and its consequences are discussed in Section 12.4.

9 Radiation Testing (Total Ionising Dose)

9.1 Rationale and Test Isolation Strategy

Crystalline MgO, the MTJ's insulating barrier, is expected to be substantially more radiation-tolerant than the amorphous SiO_2 gate oxide used in conventional silicon devices, for the reasons discussed in Section 2.2. To test this without confounding the result with unrelated failure modes, only the passive CT100 sensor itself was irradiated by itself.

9.2 Experimental Setup

Ten CT100 sensors were exposed using a 320 kV tube potential (tungsten target, 30° anode angle, 3 mm beryllium filter) at a beam current of 10 mA. The beam was seasoned (ramped up to full parameters) from 15:55 to 16:05 NZST on 22 July 2026, and held at full 320 kV/10 mA output until switch-off at 19:26 NZST, giving 3 h 21 min (12 060 s) of full-power exposure.

The physical test setup inside the irradiation chamber is shown in Figure 11. The dose rate at the sensor location was calculated as $0.0338\,\mathrm{Gy\,s^{-1}}$ (Si), equal to $3.38\,\mathrm{rad(Si)/s}$, based on the beam path approximated as 315 mm of air followed by 5 mm of aluminium attenuation before reaching the TMR sensor, itself modelled as a silicon block for absorption purposes. Applied over the 12 060 s full-power exposure window, this gives a total absorbed dose of 407.6 Gy (40.76 krad(Si)). This dose figure does not include the lower, ramping dose accumulated during the 15:55–16:05 seasoning period, so the total absorbed dose reported here is a conservative estimate.

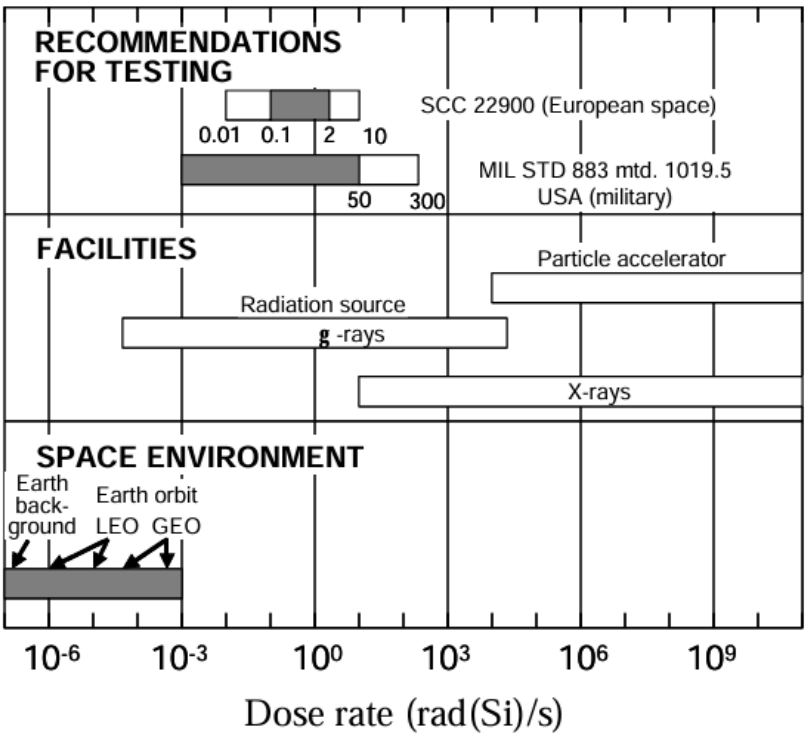


Figure 11: Comparison of the dose rates found in the radiation environments and the rates used in the testing of components for tolerance.

To contextualize the severity and relevance of this exposure, these parameters were evaluated against the radiation hardness assurance framework outlined by Poivey [29]. Actual natural space environments, such as Low Earth Orbit (LEO) and Geosynchronous Equatorial Orbit (GEO), typically expose components to extremely low, continuous dose rates below 10^{-3} rad(Si)/s. The experimental dose rate of 3.38 rad(Si)/s provides a robust accelerated testing environment that effectively simulates cumulative degradation over a practical laboratory timeframe. Notably, this rate aligns directly with the SCC 22900 (European space) recommendations, which mandate testing dose rates between approximately 0.01 and 10 rad(Si)/s[cite: 21]. Consequently, the accumulated 40.76 krad(Si) represents a fully compliant and highly relevant benchmark for standard orbital spaceflight qualification.

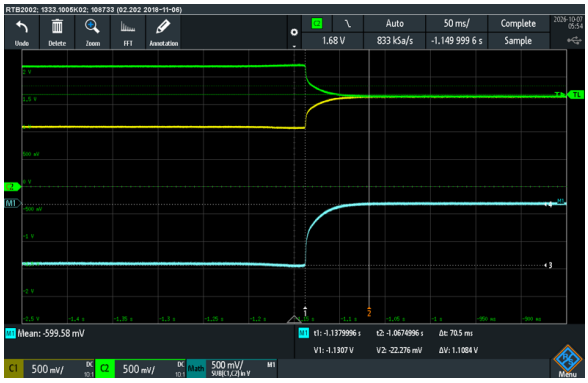
9.3 Sensor-Only Board Saturation Response Post-Irradiation

To evaluate whether radiation exposure alters sensor saturation limits or transient recovery dynamics, two irradiated units (Samples 3 and 4) were driven to magnetic saturation following the methodology established in Section 6.3. Table 5 summarizes the saturation ceilings, baseline offsets, differential swings, and recovery times across all eight trials for both non-irradiated and irradiated sample groups for comparison.

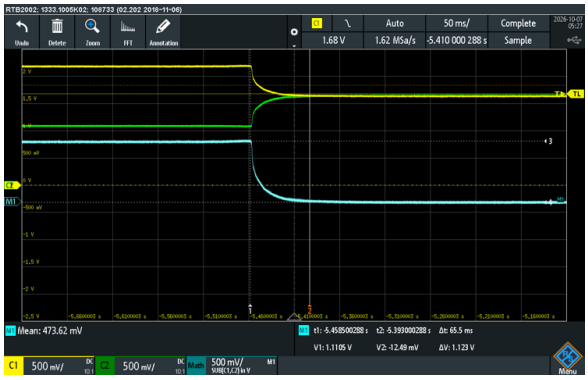
Figure 12 shows the oscilloscope captures for the irradiated samples. Following the same trace colour assignment used in the non-irradiated characterization, the blue waveform (M_1) represents the differential output voltage ($V_{\text{out}} = X_1 - X_2$), while the yellow ($C_1$) and green ($C_2$) waveforms represent the individual X_1 and X_2 bridge terminal voltages, respectively.

Table 5: Saturation ceiling and recovery time two irradiated/non-irradiated samples for comparison.

Sample	Ceiling (V)	Baseline (mV)	Swing ΔV (V)	Recovery (ms)
1	−1.126	−17.4	1.108	76.5
1	+1.125	−12.5	1.138	74.0
2	−1.145	−17.4	1.128	76.0
2	+1.125	−12.5	1.138	67.5
3	−1.131	−22.3	1.108	70.5
3	+1.111	−12.5	1.123	65.5
4	−1.145	−22.3	1.123	78.0
4	+1.111	−22.3	1.133	85.0



(a) Sample 3: Negative Saturation Recovery



(b) Sample 3: Positive Saturation Recovery



(c) Sample 4: Negative Saturation Recovery



(d) Sample 4): Positive Saturation Recovery

Figure 12: Oscilloscope captures displaying post-irradiation saturation voltage clamping and transient relaxation trajectories for irradiated sensor board Samples 3 and 4 across both field polarities.

Averaging across both polarities and units within each group, the non-irradiated group exhibits a mean saturation ceiling of 1.130 V compared to 1.124 V for the irradiated group, representing a 0.5% difference. The mean recovery time across groups is 73.5 ms non-irradiated versus 74.75 ms irradiated, a 1.7% difference.

Because the performance difference between the non-irradiated and irradiated groups averages (0.5% on saturation ceiling and 1.7% on recovery duration) falls strictly within the non-

irradiated unit-to-unit spreads, the differences observed are attributable to manufacturing variance rather than total ionizing dose (TID) degradation. Consequently, neither the saturation voltage ceiling nor the transient recovery demonstrate detectable permanent radiation-induced degradation at the time of testing. Note, both irradiated and non-irradiated samples were tested in the same session on the 7th of October 2026.

9.4 Quantitative Sweep on the Integrated Board

Following transfer to a integrated board the same current-sweep experiment as Section 8.3 was repeated results can be found in Table 6. Note, the irradated sensor had not been exposed to saturating magnetic fields prior to irradiation, both pre- and post-irradiation sweeps were performed on a bridge with zero magnetic history.

Table 6: Irradiated sample current sweep.

Current (mA)	DMM (mV)	MCU (mV)
−2999.0	−3.723	−3.258
−2595.0	−3.123	−2.803
−1999.0	−2.570	−1.748
−1512.0	−1.620	−0.044
−1017.0	−1.050	−0.001
−498.0	−0.100	0.463
−0.7	0.269	0.308
498.0	1.130	1.749
1017.0	1.640	2.657
1512.0	2.098	3.178
1999.0	2.720	3.298
2595.0	3.303	4.350
2999.0	3.912	4.834

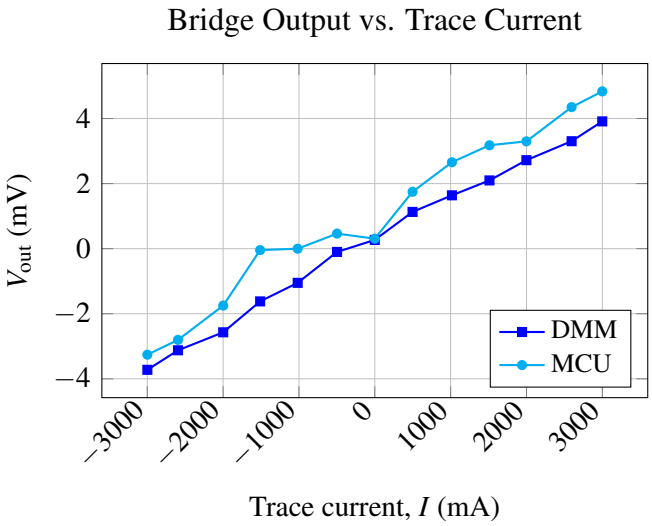


Figure 13: Plot of irradiated DMM and MCU sweeps detailed in Table 6.

The same linear-fit procedure gives a sensitivity of $1.269\,\mathrm{mV\,A^{-1}}$ on the DMM ($R^2 = 0.995$) and $1.312\,\mathrm{mV\,A^{-1}}$ on the MCU ($R^2 = 0.975$), corresponding to 93.6% (DMM) and 99.8% (MCU) of the unirradiated sensitivity (Section 8.3). Both figures are well within the level of session-to-session variation already observed on unirradiated hardware (Section 8.4), with no consistent direction of change between the two independent instruments. *No statistically significant change in sensitivity was found at the tested dose.* Given that this specific radiation configuration does not appear elsewhere in the TMR-sensor literature reviewed for this project (Section 2.2, pending confirmation), this result is offered as a preliminary but novel data point for TMR radiation tolerance under this exposure regime, supporting the broader literature consensus on MTJ radiation tolerance (Section 2.2).

9.5 Irradiated Saturation Recovery on the Integrated Board

Following the 40.76 krad(Si) Total Ionizing Dose (TID) exposure, the saturation recovery of the integrated board was evaluated with a regulator input of $V_{CC} = 5.00\,\mathrm{V}$ and a measured TMR

sensor supply of $V_{DD} = 3.289\,\mathrm{V}$, the irradiated board was subjected to the same magnetic saturation procedure as in Section 8.2. The resulting recovery waveforms are plotted in Figure 14 and the key metrics are summarized in Table 7. The following irradiated data was collected on the 9th of October 2026.

Table 7: Saturation recovery results, irradiated integrated board (Board B).

Sample	Ceiling (V)	Baseline (mV)	Swing (V)	Recovery (ms)
B	−1.146	0.0	1.146	66.7
B	+1.149	+1.6	1.147	63.2

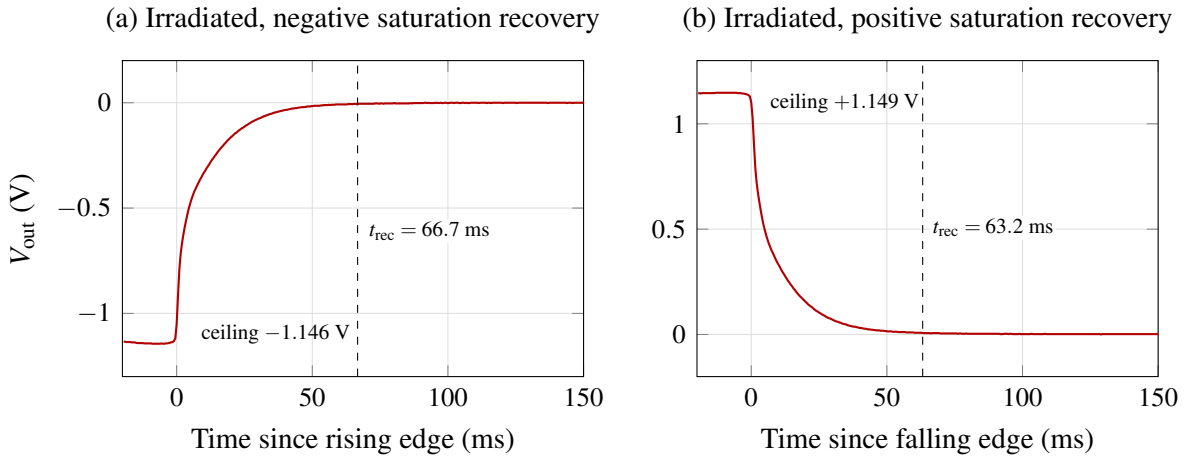


Figure 14: Irradiated Saturation recovery on the integrated board after the magnet is removed, for both field directions.

10 Magnetic Shielding

10.1 Enclosure Design

The enclosure frame was 3D printed in PLA. To check that the printed frame does not affect the field at the sensor, its relative permeability μ_r was estimated from published measurements of its magnetic susceptibility χ. The susceptibility describes how strongly a material magnetises in an applied field H where the magnetisation is given by $M = \chi H$. The flux density in the material is then:

$$B = \mu_0(H + M) = \mu_0(1 + \chi)H, \tag{22}$$

and comparing this with the definition $B = \mu_0 \mu_r H$ gives:

$$\mu_r = 1 + \chi. \tag{23}$$

Sangal et al. measured the susceptibility of common 3D printing polymers and found all of them to be weakly diamagnetic, with χ between about -8.1 and -8.7 parts per million (ppm) for PLA, PETG and ABS [30]. One part per million is 10^{-6}, so for PLA, with $\chi \approx -8.3$ ppm,

$$\mu_r = 1 + \chi = 1 + \left(-8.3 \times 10^{-6}\right) \approx 0.9999917. \tag{24}$$

This differs from the permeability of free space ($\mu_r = 1$) by less than 0.001%, so the printed frame has no measurable effect on the field at the sensor. It serves only to hold the ferrite plates and the board in place; all of the shielding comes from the ferrite and nanocrystalline layers.

Surrounding a sensor with a high-permeability material gives magnetic flux a low-reluctance path around the enclosed volume, so an external field is partly diverted away from the sensor. For an ideal closed spherical shell of inner radius a, thickness $t \ll a$ and relative permeability $\mu_r \gg 1$, the field inside is reduced by a factor of about $1 + 2\mu_r t/(3a)$ [31]. A constant ambient field adds only a fixed offset to the output, so shielding is expected to reduce that offset and any noise or drift it causes, not to change the gradient of the response to the trace current (Section 12.6).

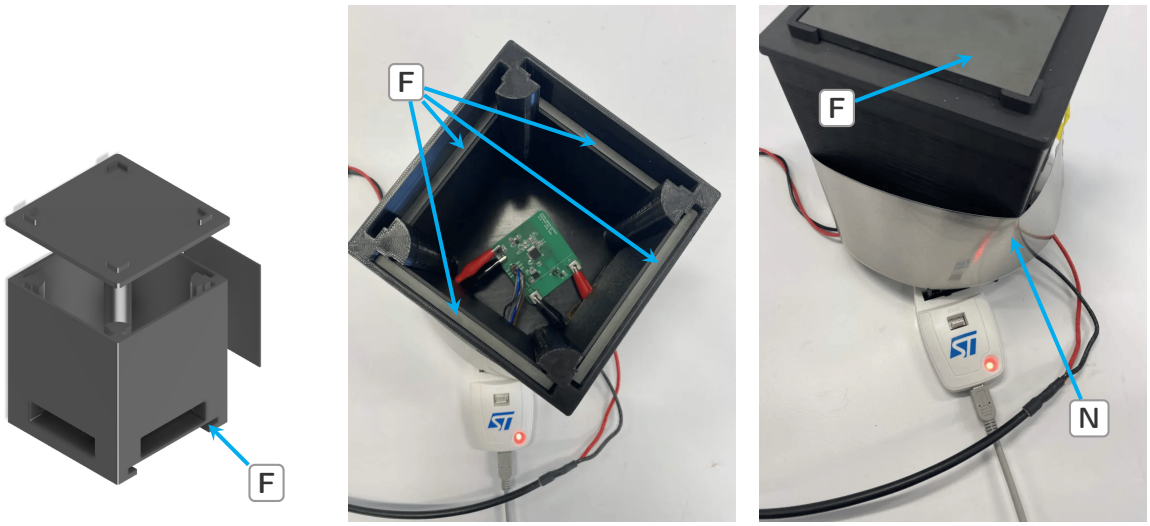
Mu-metal, an 80% nickel–iron–molybdenum alloy, is the usual choice for DC and low-frequency shielding because of its very high permeability: for fully annealed material the initial relative permeability is about 50,000 at DC and 100,000 at 60 Hz, with a maximum of about 350,000 and a saturation flux density of 0.74 T [32]. It was not used here because of its cost.

Instead, a MnZn ferrite, TDK (EPCOS) SIFERRIT material N95, was selected as a cost-effective alternative. The manufacturer quotes an initial relative permeability μ_i of 3000 at 25°C and a saturation flux density B_S of 525 mT at 25°C [33]. Six N95 slabs, each 100 mm $\times$ 100 mm $\times$ 5 mm, form the walls of the enclosure. With $\mu_r = 3000$, $t = 5$ mm and an inner radius of about 50 mm, the closed-shell expression above gives an ideal shielding factor near 200, so a closed shell would cut an external field by about two orders of magnitude. This is an upper bound: the quoted permeability is a small-signal value for a ferrite core rather than a measurement at DC in an enclosure, and the real enclosure has seams and a cable cut-out. The ambient fields to be attenuated (tens of microtesla) are far below the 525 mT saturation flux density, so the ferrite stays in its linear region.

To enable continuous operation and data logging during testing, power and data connections via an ST-Link programmer must be routed to the enclosed PCB. Consequently, a physical cut-out was designed into the enclosures to accommodate this. However, any gap effects the reluctance of the shielding and allows for ambient magnetic flux to leak and impact the TMR readings.

To reduce this flux leakage, the cut-out was covered with a flexible nanocrystalline shielding foil (YNTECH NF, Qingdao Yunlu Energy Technology). The manufacturer describes its nanocrystalline material as an Fe-based soft magnetic material with a high saturation flux density and a permeability about ten times that of ferrite [34], but gives no specific values for this foil. Measured with a digital caliper, the foil is 0.11 mm thick including its protective plastic film and about 0.02 mm thick with the film removed. The second value is close to the caliper's 0.01 mm resolution, so it is approximate.

The magnetic shielding progression—including the printed chassis slots, the internal board placement (Figure 15b), and the final exterior wrap (Figure 15c)—can be seen in Figure 15.



(a) Enclosure CAD render. (b) Internal PCB mounting. (c) Exterior shielding layer.

Figure 15: Development of the magnetic shielding enclosure. (a) The 3D-printed chassis. (b) The integrated sensor board mounted inside. (c) The assembled test configuration wrapped in shielding. Annotations indicate the placement of ferrite plates (**F**) and flexible nanocrystalline shielding (**N**).

10.2 Shielded Sweep Results

With MCU-DMM agreement already established across the full current range (Section 8), the shielded current sweep was conducted using MCU-only logging (no DMM or oscilloscope): the earlier cross-validation (Section 8) showed MCU and DMM readings agreeing to within a fraction of an ADC LSB across the unshielded sweep, so introducing the DMM into the shielded enclosure was judged to add setup complexity and added another source of possible stray magnetic field.

The non-irradiated and irradiated integrated boards was tested inside the shielded enclosure under the same sweep procedure as in Section 8. Note that the irradiated TMR sensor tested under this shielding condition had not previously been driven into saturation.

The raw measurement data for both the non-irradiated and irradiated shielded sweeps are detailed in Table 8 and Table 9, respectively. To visualize the effect of the magnetic shielding, Figure 16 compares these shielded sweeps directly against their unshielded counterparts.

Operating conditions are as follows, testing perfomed at a $V_{DD} = 3.29\,\text{V}$.

Table 8: Shielded, non-irradiated current sweep (MCU only): bridge output vs. trace current.

Current (mA)	MCU (mV)
−3043.4	8.039
−2522.5	8.549
−2057.7	9.594
−1540.3	9.949
−1040.2	11.170
−546.2	11.319
−6.3	12.705
535.4	12.946
996.2	13.931
1523.0	14.530
2038.1	15.532
2486.1	16.103
2988.7	16.474

Table 9: Irradiated, shielded current sweep (MCU only): bridge output vs. trace current.

Current (mA)	MCU (mV)
−3007.1	−3.287
−2501.2	−2.769
−2054.6	−1.130
−1567.3	0.069
−1037.7	−0.001
−532.1	0.107
−6.3	1.318
512.5	1.699
1011.4	2.837
1557.8	3.241
2048.6	3.933
2560.6	4.807
3025.4	4.932

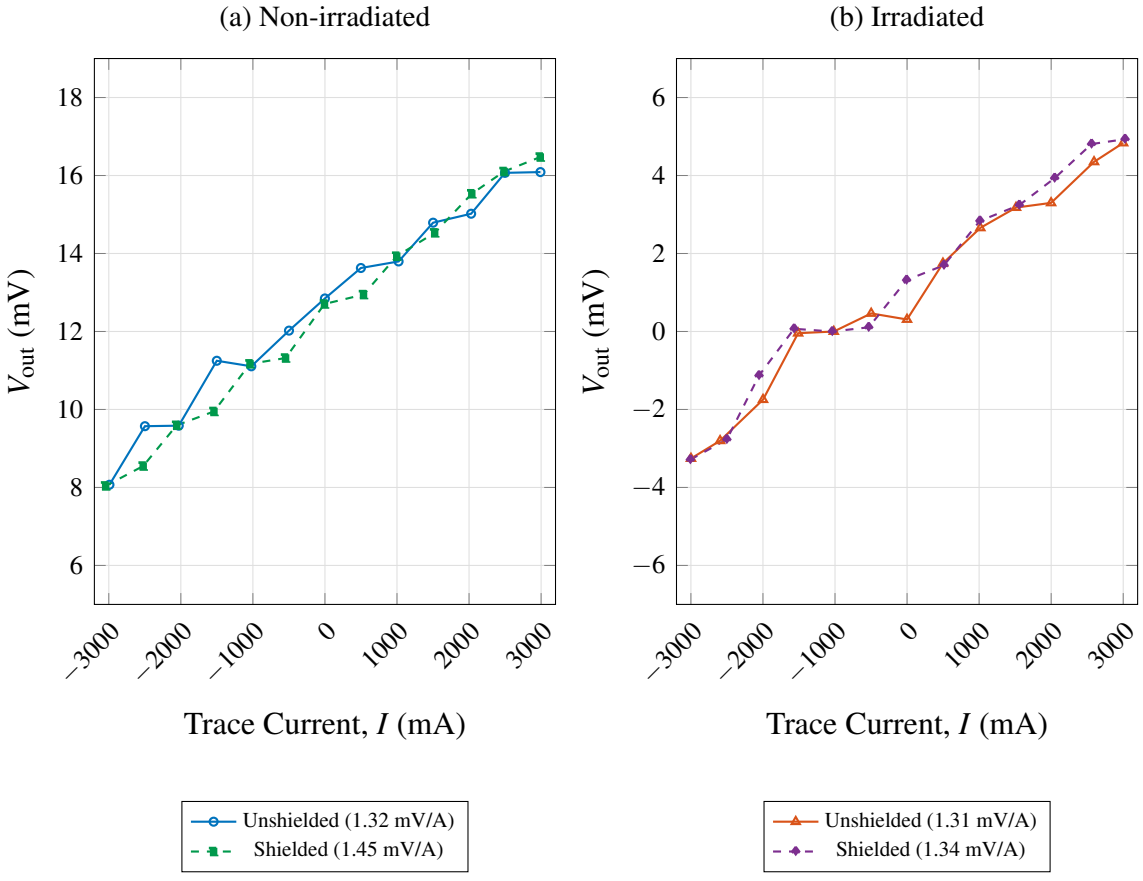


Figure 16: Shielded vs. unshielded current sweeps (MCU readings) for (a) the non-irradiated and (b) the irradiated sample.

Shielding changes the gradient by +10% on the non-irradiated board and +2.3% on the irradiated board.

10.3 Session-to-Session Variance

To better quantify the expected measurement tolerance and validate earlier findings, a secondary current sweep was performed on the non-irradiated shielded MCU board on. This validation test was conducted at $V_{DD} = 3.29\,\text{V}$.

The raw data logged for this secondary shielded sweep is detailed in Table 10. Figure 17 visualizes this data alongside the initial shielded non-irradiated sweep to illustrate the session-to-session any offset and sensitivity drift.

Table 10: Validation Sweep: Non-Irradiated Shielded Board

INA Current (mA)	MCU (mV)
−3046.47	8.039
−2492.64	8.595
−2009.76	9.604
−1493.87	10.014
−1037.81	11.110
−507.16	11.312
−4.58	12.514
494.27	12.888
1055.68	13.579
1515.83	14.455
2016.78	14.890
2487.65	15.978
3006.08	16.192

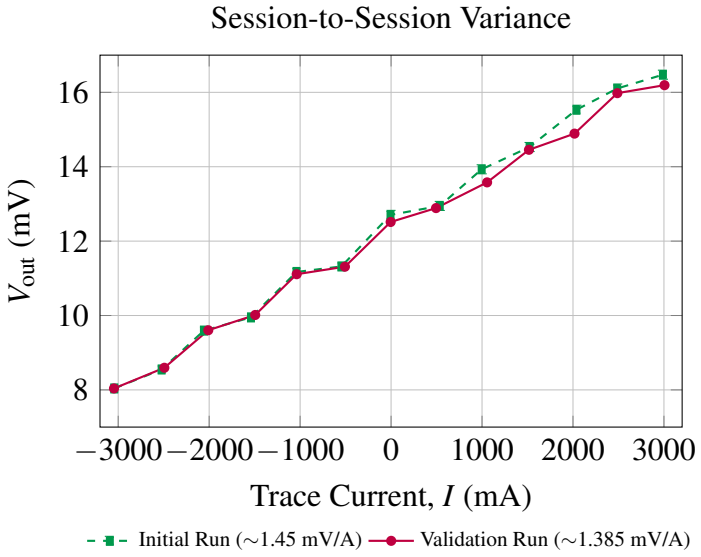


Figure 17: Plots of two discrete testing sessions on the same shielded, non-irradiated board.

A linear fit to the validation sweep gives $1.385 \pm 0.031\,\text{mV A}^{-1}$. The first shielded run on the same board gave $1.446 \pm 0.033\,\text{mV A}^{-1}$ (Section 10.2), a difference of 4.2% between two sessions under the same conditions, about 1.3 times their combined standard error. This is the session-to-session variation of the method.

This variation is the yardstick for the shielding comparison. On the non-irradiated board the change from unshielded to shielded is +5% or +10% depending on which shielded run is used, only slightly larger than the session-to-session variation, and on the irradiated board it is +2.3%, smaller than it. The data therefore do not demonstrate a clear shielding effect.

11 Quantitative Magnetic Hysteresis Testing

To isolate the effects of magnetic history from potential board-level hysteresis, the sensor on the non-irradiated integrated board was desoldered and replaced with a new sensor that had never been exposed to a magnetic field (Board C). All measurements were taken on 12 October 2026 with a regulator input of $V_{CC} = 5.00\,\text{V}$ and a sensor supply of $V_{DD} = 3.321\,\text{V}$.

The test was run in one session, with no other change to the board, in three steps: (i) a current sweep on the untouched sensor before saturation; (ii) saturation to the negative ceiling and then to the positive ceiling; and (iii) a second, identical current sweep after saturation. The two

sweeps are listed in Tables 11 and 12 and plotted in Figure 18. Panel (a) of that figure shows the raw output of both sweeps, and panel (b) shows them again with each sweep's fitted offset removed so that only the gradient is compared. The saturation events themselves are reported in Table 13 and Figure 19.

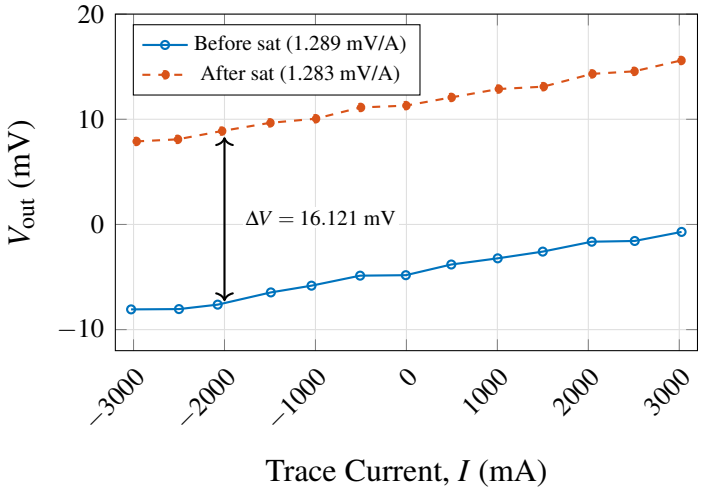
Table 11: Before Saturation

INA (mA)	MCU (mV)
−3026.78	−8.076
−2500.36	−8.047
−2074.76	−7.625
−1490.19	−6.462
−1043.62	−5.821
−508.79	−4.869
−6.30	−4.824
493.28	−3.809
1003.71	−3.222
1498.85	−2.582
2037.68	−1.641
2510.09	−1.571
3025.05	−0.709

Table 12: After Saturation

INA (mA)	MCU (mV)
−2971.20	7.896
−2516.89	8.091
−2031.67	8.878
−1500.70	9.660
−999.23	10.051
−504.66	11.124
−6.15	11.297
492.07	12.077
1010.23	12.875
1504.07	13.100
2039.39	14.316
2502.52	14.558
3018.47	15.590

(a) Raw output



(b) Offset removed

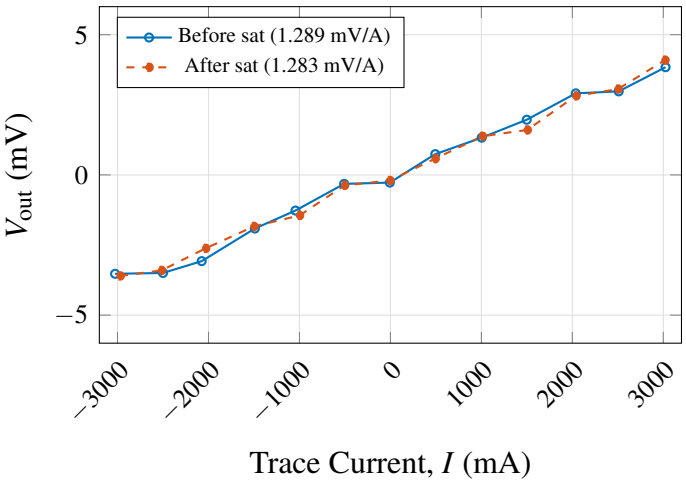


Figure 18: Current sweep on a fresh, never-saturated sensor before (solid) and after (dashed) being driven to saturation. (a) Raw output: the curve moves up by 16.121 mV. (b) With offset removed, sweeps overlap.

Table 13: Saturation recovery results, fresh sensor on the integrated board (Board C).

Sample	Ceiling (V)	Baseline (mV)	Swing (V)	Recovery (ms)
C	−1.107	−1.6	1.105	40.1
C	+1.117	+11.3	1.105	48.7

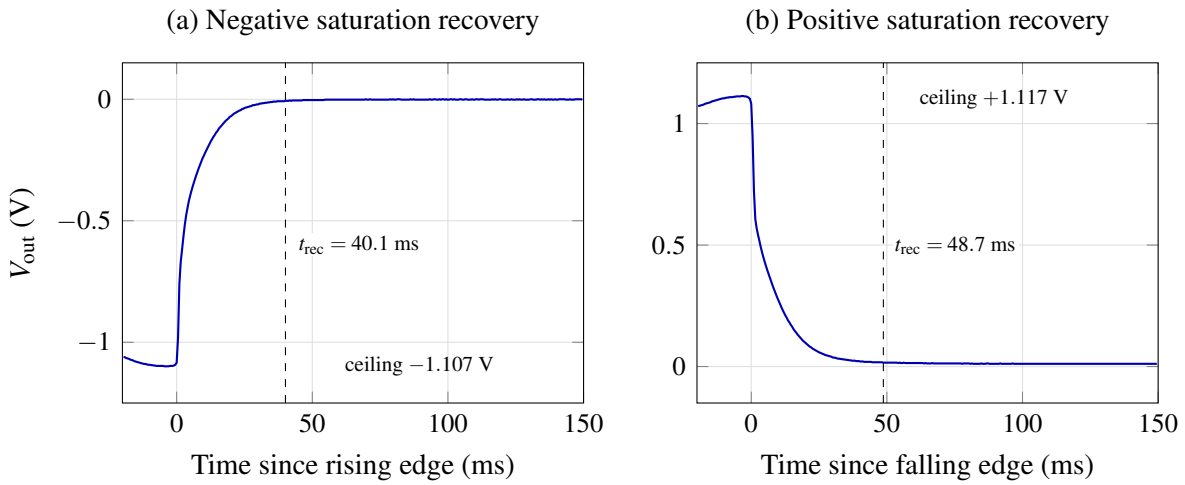


Figure 19: Newly Soldered Non-Irradiated Saturation recovery on the integrated board after the magnet is removed, for both field directions.

When the saturating magnetic field is removed, the sensor output decays from its respective ceiling (-1.107 V for negative saturation, 1.117 V for positive saturation). Figure 19 tracks this transient response, showing that the sensor settles to within 5 mV of its new baseline in 40.1 ms after the negative event, and 48.7 ms after the positive event.

The zero-field baseline is the output once the field has been removed. Table 14 lists the baseline before and after each saturation event for all three integrated boards, and Figure 20 shows the baseline of Board C through the sequence. “Before” is the median output in a quiet interval before the field was applied, “after” is the median output once recovery was complete, and the shift is after minus before, so a positive shift means the zero-field output moved up. The shifts are computed from unrounded values.

Table 14: Baseline before and after each saturation event, integrated boards.

Sample	Ceiling reached	Before (mV)	After (mV)	Shift (mV)
A (non-irradiated)	Negative	−6.4	−8.1	−1.6
A (non-irradiated)	Positive	−8.1	+12.9	+20.9
B (irradiated)	Negative	0.0	0.0	0.0
B (irradiated)	Positive	0.0	+1.6	+1.6
C (fresh sensor)	Negative	−4.8	−1.6	+3.2
C (fresh sensor)	Positive	−1.6	+11.3	+12.9

Three observations follow from Table 14. First, on Board C the negative ceiling moved the baseline by only +3.2 mV (two ADC steps of 1.61 mV), but the positive ceiling that followed moved it by +12.9 mV, leaving the baseline at +11.3 mV, a total shift of +16.1 mV from the untouched sensor. Second, Board A shows the same pattern, a negligible shift at the negative ceiling and a large one (+20.9 mV) at the positive ceiling. Third, the irradiated Board B shows no comparable shift (0.0 and +1.6 mV). Because the negative ceiling was always reached first, these data cannot yet separate the direction of the field from the order of the events, and Board B is a single irradiated unit.

[TODO: Planned measurement: repeat this test on a second fresh sensor with the positive

ceiling reached first, using the same session procedure. Add the two rows to Table 14 and state whether the large shift follows the field direction (positive ceiling) or the order of the events (second saturation).]

Gradient stability. Despite the shift in offset, the sensor's linear response to current is essentially unaffected. A straight-line fit to each sweep gives a gradient of 1.289 ± 0.038 mV/A before saturation and 1.283 ± 0.030 mV/A after, a change of only -0.5%, which is far smaller than the standard error of either gradient. The fitted offset at zero current moves from -4.55 mV to $+11.50$ mV, a shift of 16.05 mV, and R^2 stays above 0.99 (0.991 before and 0.994 after). Saturation therefore moves the whole response up without changing its slope or its linearity. This is the pattern seen in panel (b) of Figure 18, where the two sweeps overlap once their offsets are removed.

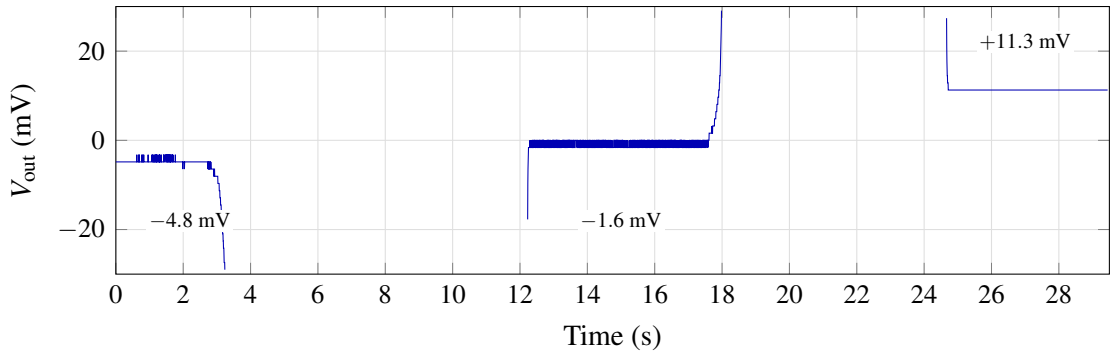


Figure 20: Output baseline before, between and after the two saturation events (view clipped to ± 30 mV so the baseline is visible). The baseline moves from -4.8 mV to -1.6 mV after the negative saturation, then to $+11.3$ mV after the positive saturation.

12 Discussion

This project asked whether a low-cost commercial TMR sensor, the CT100, is operationally viable for CubeSat ADCS/EPS use, focusing on how it behaves when driven into saturation and after irradiation as conducted in Section 3. The motivation (Section 1) is that CubeSat magnetometers must be small, light and low power, and that TMR offers high sensitivity in a few-millimetre package but has little space heritage (Section 2.1). This section first checks that the test method can be trusted, then discusses each result against the literature and the CubeSat use case, and ends with the limitations of the work.

Table 15 summarises how the results answer the hypotheses of Section 3.1, and the subsections that follow discuss each one.

Table 15: Outcome of each hypothesis.

Hypothesis	Outcome	Evidence
H1: Ceiling at $\pm V_{DD}/3$	Supported	Every ceiling within about 5% of $V_{DD}/3$; positive and negative ceilings differ by 0.9 to 2.4%.
H2: Gradient matches model	Supported	All nine fits lie within -5% to $+9\%$ of the predicted $1.332\,\mathrm{mV\,A^{-1}}$, equivalent to 4.3 to $4.9\,\mathrm{mV\,V^{-1}\,mT^{-1}}$, inside the datasheet range of 3.8 to 5.2.
H3: Recovery within one 10 Hz sample	Partly supported	All recoveries (40 to 85 ms) are shorter than 100 ms, but the guard window with a safety factor (127.5 to 170 ms) is not, so 1 to 2 samples must be discarded.
H4: No lasting effect of saturation	Rejected	Baseline shifted by 16.1 mV, ten ADC steps; gradient changed by only -0.5%.
H5: No radiation effect	Supported	Ceiling (0.5%), recovery (1.7%) and gradient (93.6 to 99.8%) within unit and session variation; small sample.
H6: Shielding changes response	Not supported	Gradient change of $+5$ to $+10\%$ (non-irradiated) and $+2.3\%$ (irradiated) against 4% session variation and 2 to 5% fit standard errors; the test was insensitive to an offset-only effect.

12.1 Validity of the Test Method

The results depend on the measurement chain, so it was validated before it was used. The MCU voltage readings agree with the DMM to within 0.9 mV, less than one ADC step, and the INA219 current reading agrees with the DMM to within about 1 to 2% (Section 8.4). The oscilloscope disagreed with both for DC measurements and was therefore used only for transients, where its time resolution matters. On that basis the MCU alone was used for the shielded sweeps and the hysteresis test, which saved a large amount of test time, while the radiation sweeps used both the DMM and the MCU so that they were directly comparable with the earlier baseline. Two further checks support the method. The field model, built only from the board stack-up and the datasheet position of the sensing element, predicts the measured gradients to within a few percent (Section 8.3), and repeated sweeps on the same board agree within about 4% (Section 10.3). The method is therefore good enough for the effects reported here, which are offset shifts of 13 to 21 mV and gradient changes of a few percent. It is not good enough for signals at the level of Earth's field, as discussed in Section 12.8.

12.2 Field Model and Sensitivity

Across the seven MCU sweeps the measured gradient lies between 1.283 and $1.446\,\mathrm{mV\,A^{-1}}$ (mean $1.33\,\mathrm{mV\,A^{-1}}$), and the two DMM sweeps give 1.269 and $1.356\,\mathrm{mV\,A^{-1}}$ (Table 16). The field model predicts $1.332\,\mathrm{mV\,A^{-1}}$, so every fit lies within -5% to $+9\%$ of the prediction and the mean agrees to better than 1% (Figure 21). Expressed as sensitivities, the fits give 4.3 to $4.9\,\mathrm{mV\,V^{-1}\,mT^{-1}}$, around the datasheet typical value of 4.5 and well inside its range of 3.8 to

5.2 [21]. Figure 22 compares the four test conditions with each sweep's offset removed: all four lie on the theoretical line within their scatter.

This agreement depends on getting the geometry right. The model is most sensitive to the distance d between the trace and the sensing element: a model that ignored the height of the sensing element inside its package ($d \approx 0.42\,\mathrm{mm}$) would predict a gradient about 70% higher than measured. With the datasheet sensor position, the remaining spread between sweeps (up to about 9%) is comparable to the repeatability of one board (4%, Section 10.3), the standard error of each fit (2 to 5%) and the tolerance on the sensor height (-5% to $+7\%$ in the predicted gradient). A constant ambient field cannot explain any of it, because it adds an offset without changing the gradient. For a CubeSat designer the result is that the datasheet sensitivity and a simple strip-conductor model are good enough to design a test or calibration coil to within about 10%, although each flight unit would still be calibrated.

Table 16: Linear fit of every current sweep: gradient with standard error, offset and R^2.

Sweep	Gradient (mV/A)	Offset (mV)	R^2
Non-irradiated, unshielded (DMM)	1.356 ± 0.067	13.09	0.974
Non-irradiated, unshielded (MCU)	1.315 ± 0.055	12.60	0.981
Irradiated, unshielded (DMM)	1.269 ± 0.026	0.22	0.995
Irradiated, unshielded (MCU)	1.312 ± 0.063	1.00	0.975
Non-irradiated, shielded (MCU)	1.446 ± 0.033	12.39	0.994
Non-irradiated, shielded repeat (MCU)	1.385 ± 0.031	12.25	0.995
Irradiated, shielded (MCU)	1.342 ± 0.065	1.21	0.975
Fresh sensor, before saturation (MCU)	1.289 ± 0.038	-4.55	0.991
Fresh sensor, after saturation (MCU)	1.283 ± 0.030	11.50	0.994

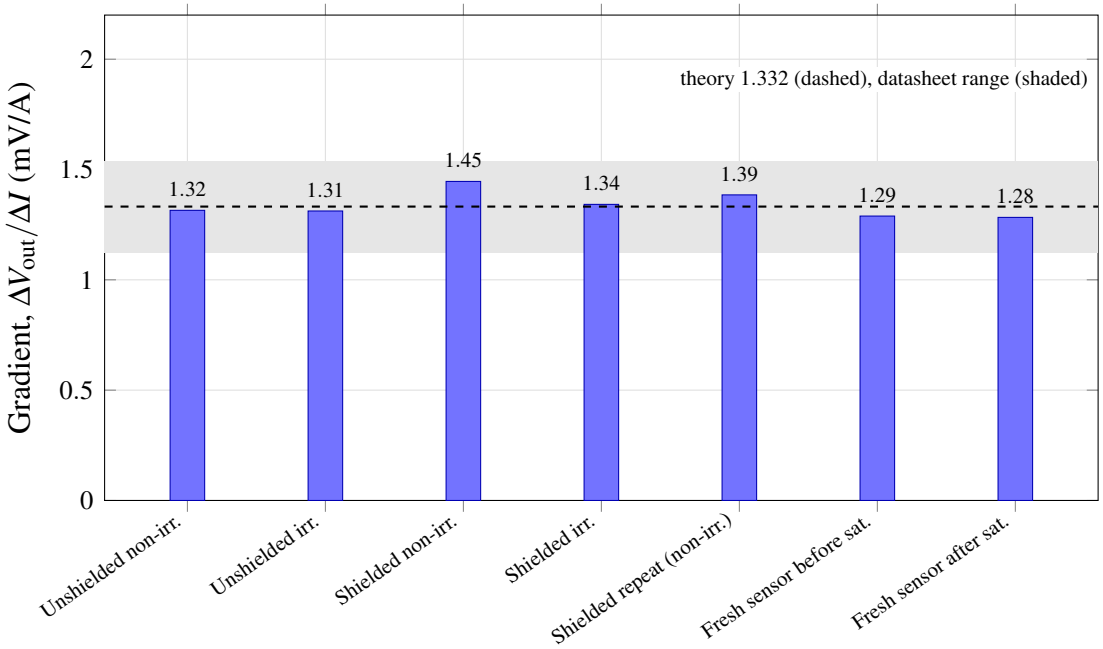


Figure 21: Measured current-to-output gradient for all seven current sweeps (MCU readings), against the theoretical gradient (dashed line) and the range implied by the datasheet sensitivity of 3.8 to $5.2\,\mathrm{mV\,V^{-1}\,mT^{-1}}$ at $V_{\mathrm{DD}} = 3.32\,\mathrm{V}$ (shaded).

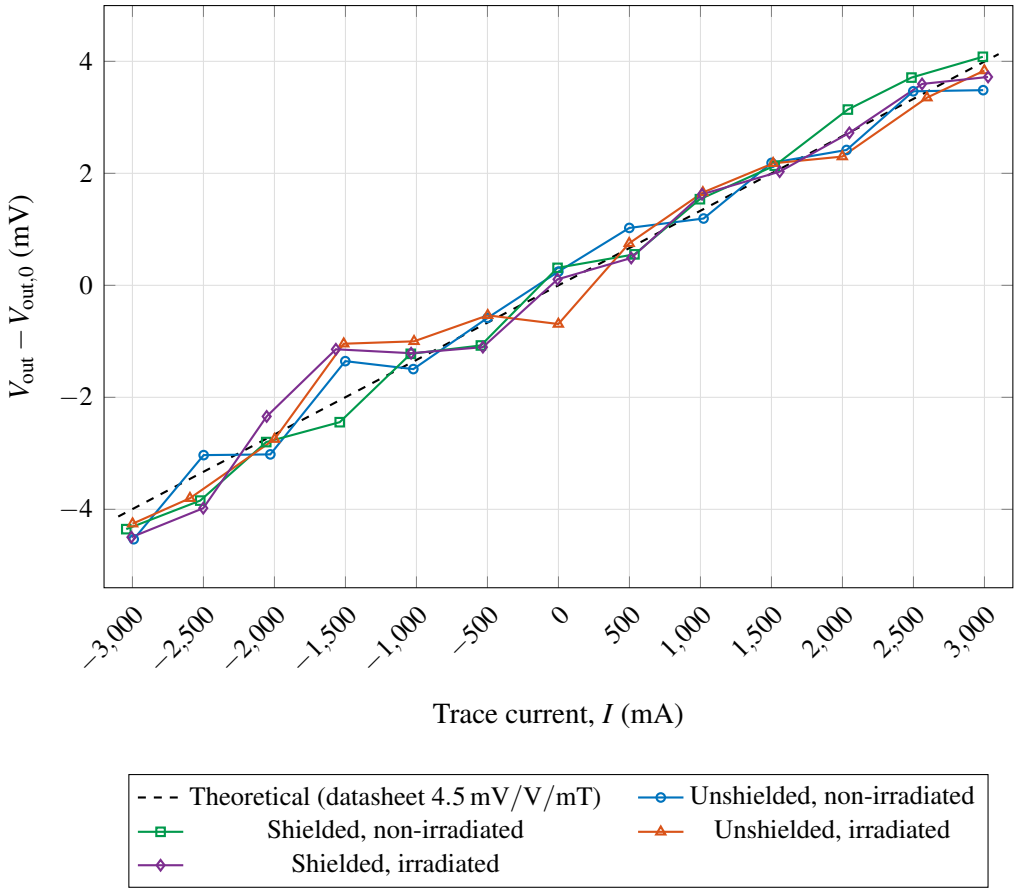


Figure 22: Current sweeps for the four test conditions with each sweep's fitted zero-current offset removed, so that only the gradient is compared, against the theoretical gradient (dashed). All measured sweeps lie close to the theoretical line.

12.3 Saturation Recovery and What It Means for a CubeSat

Table 17 summarises the saturation recovery across all samples. The ceiling is within a few percent of $V_{DD}/3$ on every unit, confirming the bridge model of Section 4.3, and it is symmetric about zero. Recovery takes 40 to 85 ms depending on the unit, and on the three integrated boards the decay is a clean exponential with a time constant of 9 to 14 ms, with no difference between the non-irradiated and irradiated boards.

Table 17: Summary of saturation recovery across all samples (means over both ceilings).

Group	Units	Ceiling (V)	Recovery (ms)	τ (ms)
Sensor-only, non-irradiated (Samples 1–2)	2	1.130	73.5	—
Sensor-only, irradiated (Samples 3–4)	2	1.124	74.8	—
Integrated, non-irradiated (Board A)	1	1.145	62.7	13.2
Integrated, irradiated (Board B)	1	1.148	65.0	13.3
Integrated, fresh sensor (Board C)	1	1.112	44.4	9.1

What this means depends on how often the on-board computer reads the magnetometer. Published CubeSat designs read magnetometers at rates from about 1 Hz, the rate of the whole

attitude-control loop on InnoCube [35], up to about 10 Hz for a commercial CubeSat magnetometer [17]. At 10 Hz the sample interval is $T_s = 100$ ms, so the worst measured recovery time of 85 ms (Sample 4) leaves a margin of only 15 ms. Comparing the recovery *frequency* ($1/t_{\mathrm{rec}} \approx 12$ to 25 Hz) with the sampling frequency is misleading, because a saturating event is not synchronised with the sampling clock. In the worst case the event ends just before a scheduled sample, and that sample is taken while the sensor is still recovering. An unrecovered reading would appear to a state estimator as a sudden impulse in the field and could trigger a false correction. The design quantity is therefore a guard window that covers the worst-case recovery with margin. Applying a conventional safety factor of 1.5 to 2 to the worst measured recovery gives a guard window of 127.5 to 170 ms:

$$T_{\mathrm{guard}} = (1.5 \text{ to } 2) \times t_{\mathrm{rec,max}} = (1.5 \text{ to } 2) \times 85\,\mathrm{ms} \approx 127.5 \text{ to } 170\,\mathrm{ms}. \tag{25}$$

This exceeds one sample interval at 10 Hz, so the first one or two readings after a saturating event should be discarded. At a 1 Hz loop the recovery is over long before the next sample and no blanking is needed. At sample rates of hundreds of hertz, such as the 500 Hz maximum configurable rate of one commercial magnetometer [36], tens of consecutive samples would be affected, so a longer blanking period, or fusion with another attitude sensor during recovery, would be required.

Saturation is easy to detect in flight, because the output goes to a ceiling of about $\pm V_{\mathrm{DD}}/3$ (± 1.1 V), which is more than a thousand times larger than the output for Earth's field. Flight software can therefore recognise a saturating event from the reading itself and then apply the guard window. The sources of such events are off-nominal (Section 2.3): a normally operating magnetorquer of about $0.2\,\mathrm{A\,m^2}$ gives a few millitesla at a few centimetres [18], which would shift the output by tens of millivolts but is well inside the CT100's ± 50 mT range. Saturation would instead come from a much closer or stronger source, such as a magnet near the sensor or magnetising equipment during ground handling. Stress-testing saturation is still worthwhile because it characterises the worst case the sensor must survive and recover from, regardless of which source causes it.

The measured recovery times should be treated with caution because the magnet was withdrawn by hand. The datasheet bandwidth of 1 kHz [23] implies an electrical response time of well under a millisecond, so the 9 to 14 ms time constants are not set by the sensor's bandwidth. They come either from the speed at which the field was removed or from the magnetic relaxation of the sensor, and the present measurement cannot separate the two (Section 12.8).

[TODO: Planned measurement: repeat the saturation recovery five times on one integrated board in one session, with the same magnet and approach, and report the recovery time and time constant as mean ± standard deviation in Table 17.]

12.4 Offset Shift After Saturation

The most significant finding of this work is that saturation moves the zero-field output without changing the gradient. On a fresh sensor with no magnetic history, two saturation events shifted the baseline by 16.1 mV (from -4.8 to $+11.3$ mV), while the gradient changed by only -0.5% and the response stayed linear (Section 11). Board A showed a similar shift of $+20.9$ mV at the positive ceiling. At the sensor's sensitivity of about $14.9\,\mathrm{mV\,mT^{-1}}$ a shift of 16 to 21 mV is equivalent to a field of 1.1 to 1.4 mT, which is at least 15 times Earth's field of 25 to 65 µT [20]. This is consistent with the manufacturer's statement that a strong field can leave a stable,

irreversible offset shift and that recalibration is recommended [22], and it puts a size on the effect for this part. The datasheet offset specification of ±5 mV/V (±16.6 mV at $V_{DD} = 3.32$ V) [21] would still be met after the shift, so the part would not be flagged as out of specification even though the shift is large compared with the geomagnetic signal.

Because the gradient and linearity are unchanged, the shift behaves like a movement of the operating point, as expected if the free layer relaxes to a slightly different resting state after saturation. For a CubeSat this means that a one-point re-zeroing of the offset is sufficient after a saturating event, and that a full recalibration is not needed. The saturation can be detected from the reading itself (Section 12.3), so the recommended procedure is to detect the ceiling, discard the guard window, and then re-estimate the offset before trusting the magnetometer again. In orbit the sensor is never in zero field, so the offset cannot simply be measured; it would instead be re-estimated by attitude-independent calibration against a geomagnetic field model, as demonstrated in flight for the MAGIC magnetometer on CINEMA [10].

This also explains the difference in baseline between the non-irradiated and irradiated sweeps (about +12 mV against about +1 mV). The non-irradiated board had been saturated before its sweeps, the irradiated board had not, and the fresh sensor shows that saturation alone moves the baseline by this amount. It is therefore not evidence of a radiation effect on the offset. One point remains open: Board B's own saturation events moved its baseline by only 1.6 mV, so it is not yet known whether irradiated sensors respond to saturation differently. The evidence is also limited by the order of the events, because in every run the negative ceiling was reached before the positive one, so the direction of the field and the order of the events cannot yet be separated.

12.5 Radiation Tolerance

Ten sensors were exposed to 40.76 krad(Si), and three were tested (two on sensor-only boards and one on the integrated board). No change was found in the saturation ceiling (0.5% difference), the recovery time (1.7%), the recovery time constant (13.3 against 13.2 ms) or the gradient (93.6 to 99.8% of the non-irradiated value, within session-to-session variation). This is what the literature predicts: MTJ sensors have been shown to tolerate 5 Mrad(Si) [11] because the crystalline MgO barrier has far fewer trap sites for radiation-induced charge than the amorphous oxides in CMOS devices [15]. The dose used here is modest by comparison with that work and with the 300 krad(Si) used for a COTS magnetometer [16], but it is above the 10 krad component-level dose quoted for a commercial CubeSat magnetometer [17]. The contribution is device-level data for one low-cost packaged part, including, to the author's knowledge, saturation recovery after irradiation. The test has several limits. The sensors were not monitored during exposure, so transient effects that vanish afterwards were not seen, and the bias condition during exposure needs to be stated (Section 9.2). X-rays test ionising dose only, not displacement damage or single-event effects. The comparison is between different units with different magnetic histories, with two sensor-only samples and one integrated board per group. The result should therefore be read as a first indication that permanent radiation damage at this dose is small, not as a qualification.

12.6 Magnetic Shielding

Shielding did not measurably change the response. The gradient changed by +5 to +10% on the non-irradiated board and +2.3% on the irradiated board (Figure 16), and a repeat shielded

sweep differed from the first by 4% (Section 10.3), so the shielding effect is at most comparable to the session-to-session variation and the 2 to 5% standard error of each fit. This should not be read as proof that the shield does not work, for three reasons. First, a constant ambient field changes the offset and not the gradient, so a gradient comparison is an indirect test of shielding at best. Second, Earth's field of tens of microtesla is only about 0.7 mV at the CT100's sensitivity, less than one ADC step of 1.61 mV, so even a perfect shield would not have produced a visible change with this measurement chain. Third, the enclosure is far from ideal: a closed shell of the N95 ferrite would at best reduce an external field by a factor of about 200 (Section 10.1), and the real enclosure has seams and a cable cut-out. The ferrite close to the trace may also change the field of the trace itself, which could account for the +10% gradient change on one board. A proper test would apply a known external field with and without the enclosure and compare the zero-current baseline and noise on a higher-resolution measurement. Mu-metal could then be tried if the ferrite proves insufficient. For flight, magnetometer placement away from sources matters more than shielding the sensor [17].

12.7 Power Consumption

Power is a first-order constraint on a CubeSat, so the cost of the sensing element is relevant. The datasheet gives a typical power consumption of 0.30 mW at $V_{DD} = 3.0$ V with a 30 kΩ bridge [21]. Since the bridge draws V_{DD}/R, where R is the bridge resistance, this scales to about 0.36 mW at the 3.30 V used here, and to 0.27 to 0.54 mW over the 20 to 40 kΩ resistance range. It falls with the square of the supply voltage down to the datasheet minimum of 1.0 V. For comparison, the AMR-based MAGIC instrument draws 0.5 W [4] and a commercial CubeSat magnetometer about 0.7 W [17]. These are complete instruments with electronics, so the comparison is not like for like, but it shows that the sensing element itself is a negligible part of a magnetometer's power budget; the power of a full instrument is set by the gain stage, ADC and processor. **[TODO: Planned measurement: connect the DMM in series with the supply to the standalone sensor board (DMM on its microamp or milliamp current range), record the supply current at zero field and with the sensor held at saturation, and report the current and the power $P = V_{DD}I$ here in place of the datasheet figure.]**

12.8 Limitations and Better Ways to Measure the Response

The main weakness of the current method is the saturating stimulus. A hand-held magnet gives an uncontrolled withdrawal speed, an unknown field at the sensor, and poor repeatability, so recovery times include the operator and cannot be compared closely between units. A better method would use an electromagnet driven through a fast switch, so that the field is removed in a controlled, repeatable way, with the coil current and the sensor output recorded together on the oscilloscope and several repeats taken to give a mean and standard deviation. A linear actuator moving a magnet at a fixed speed would also work. The trace used for the current sweeps cannot saturate the sensor, because a field of about 20 mT would need on the order of 180 A.

Other limitations are as follows.

- **Resolution.** With an unamplified 12-bit ADC, one step of 1.61 mV is about 108 µT at the CT100's sensitivity, larger than Earth's field. The sensor itself is far quieter: its datasheet output noise of 700 nV/$\sqrt{\text{Hz}}$ at 10 Hz ($V_{DD} = 1.0$ V) [21] corresponds to about

$0.16\,\mu T/\sqrt{Hz}$, or roughly 0.5 µT RMS over a 10 Hz bandwidth, about 1% of Earth's field. The readout, not the sensor, therefore limits resolution in this work. A gain stage (whose slew rate would then have to be checked against the recovery measurement) or a higher-resolution ADC would be needed for geomagnetic sensing. **[TODO: Planned measurement: record 60 s of zero-field output with the DMM on its slowest, most precise setting, report the standard deviation, and convert it to a field with the $14.9\,mV\,mT^{-1}$ sensitivity.]**

- **Oscilloscope resolution.** At 500 mV/div the oscilloscope steps are about 5 mV, so the sensor-only baselines are not interpretable and the ceilings carry about ±5 mV of uncertainty.
- **Sample size and design.** There are two sensor-only samples per group, one board per group on the integrated board, a single dose level, and the irradiated and non-irradiated comparison is across different units.
- **Hysteresis test.** The two sensors tested were saturated negative first, and only one fresh sensor was tested in a controlled before-and-after sequence.
- **Field model.** The predicted gradient carries a −5% to +7% uncertainty from the tolerance on the sensor's height inside its package and the package stand-off, and the Ansys model still needs to be evaluated at the datasheet sensor height (Section 7.4).
- **Scope of the project.** Temperature effects were studied separately by the project partner and are not covered here.

13 Conclusions and Future Work

Conclusions. This project characterised a low-cost commercial TMR sensor, the CT100, using two purpose-built PCBs, a field model built from the board stack-up and the datasheet sensor position, and instruments that were validated against each other. For the conditions tested, the research question of Section 3 is answered as follows.

- **Bridge behaviour.** The saturation ceiling is within about 5% of $\pm V_{DD}/3$ on every unit tested and symmetric about zero, as derived from the bridge topology (H1 supported).
- **Sensitivity.** The measured current-to-output gradient agrees with the field model within −5% to +9%, with the mean within 1%, equivalent to 4.3 to $4.9\,mV\,V^{-1}\,mT^{-1}$ against a typical datasheet value of 4.5. The response is linear, with R^2 of 0.97 or higher (H2 supported).
- **Recovery.** Recovery from saturation takes 40 to 85 ms, as an exponential decay with a time constant of about 9 to 14 ms. At a 10 Hz magnetometer rate the first one to two samples after a saturating event must be discarded; at 1 Hz no action is needed (H3 partly supported).
- **Lasting effect of saturation.** Saturation shifts the zero-field offset by about 13 to 21 mV (1 to 1.4 mT equivalent) without changing the gradient or the linearity, so the offset must be re-estimated after a saturating event (H4 rejected). This matches the manufacturer's note on offset shift and puts a size on it for the CT100.

- **Radiation.** No change attributable to radiation was found in the ceiling, recovery, time constant or gradient after 40.76 krad(Si), as expected from the literature on MTJ sensors (H5 supported, small sample).
- **Shielding.** Ferrite shielding did not measurably change the response (H6 not supported). The test was insensitive to an offset-only effect, so shielding effectiveness remains unresolved.
- **Power and resolution.** The bridge consumes about 0.3 to 0.5 mW, a negligible part of a magnetometer power budget. The sensor's intrinsic noise of about $0.16\,\mu\mathrm{T}/\sqrt{\mathrm{Hz}}$ would resolve Earth's field, but the unamplified 12-bit readout used here could not.

Significance. The CT100 is a promising low-cost COTS candidate for CubeSat magnetometry. It tolerates ionising radiation at the dose tested, survives saturation, recovers within tens of milliseconds and keeps its gradient. Its one lasting change, a shift in its zero point, can be detected from the reading itself and corrected by the kind of in-flight offset calibration already used for chip-scale magnetometers [10]. The work also provides a reusable, low-cost test method, from a PCB trace field source modelled from the stack-up to microcontroller logging validated against a DMM, that can be applied to other COTS TMR parts. More units, more dose levels and a readout with sufficient resolution are needed before flight use.

Future work.

- Replace the hand-held magnet with a controlled, repeatable saturation source (an electromagnet with a fast switch) and record several repeats per sample.
- Repeat the hysteresis test on several fresh sensors, with the positive ceiling first as well as the negative, and on a fresh irradiated sensor, and implement and test a firmware procedure that detects the ceiling, applies the guard window and re-estimates the offset.
- Monitor sensors during irradiation, state the bias condition, and extend the tests to more dose levels and to proton irradiation.
- Test shielding with a known external field and a higher-resolution measurement, and try mu-metal.
- Add a gain stage or a higher-resolution ADC and measure the noise floor in field units.
- Re-run the Ansys model at the datasheet sensor height, and combine these results with the thermal-cycling results of the project partner.

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